

How Resilient Is Mortgage Credit Supply? Evidence from the COVID-19 Pandemic

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ABSTRACT

We study the resilience of U.S. mortgage credit supply during the COVID-19 pandemic—the most significant shock since the financial crisis—and draw out broader lessons about the functioning of this important market. While mortgage lending boomed in 2020 and 2021, we find that a sharp increase in intermediation markups limited the pass-through of low rates to households. We link this increase in markups to capacity constraints amplified by pandemic-related operational and labor market frictions. We also present new evidence that capacity constraints in the mortgage market are national in scope and have not yet been significantly mitigated by recent technological change. Nonbank lenders, often thought to be fragile, gained market share from banks but remain reliant on securitization. We also find evidence that government credit guarantees support the flow of credit to risky borrowers but are not always sufficient, and that quantitative easing particularly boosts credit supply for the specific types of loans being purchased.

The pandemic years 2020 and 2021 were an extraordinary period for the U.S. mortgage market. In both years, lenders underwrote over \$4 trillion in new mortgages, far higher than in any other year except 2003.¹ Rates on 30-year fixed-rate mortgages (FRMs) fell to historic lows below 3% for an extended period. At the same time, however, the pandemic was of course also a large negative economic shock with effects on borrower default risk and financial markets more broadly, and virus-related restrictions complicated many activities including housing transactions and the origination of mortgages.

We use this period to study the resilience of U.S. mortgage credit supply, focusing on broader lessons for understanding the functioning of this critical market. One key goal is to shed light on the effects of well-documented structural changes in the mortgage market since the 2007 to 2009 global financial crisis (GFC), including a secular shift in lending to less-regulated nonbanks (Buchak et al., 2018; Kim et al., 2022), the diffusion of online lending (Buchak et al., 2018; Fuster et al., 2019) and changes in regulation and the role of government. These developments raise many questions: How resilient are the new nonbank lenders in periods of stress? Has new technology alleviated lender capacity constraints, making credit supply more elastic? How sensitive are mortgage risk premia to economic and financial shocks, and what role do government guarantees and other interventions (e.g., quantitative easing, or QE) play in mitigating these effects?

We start by documenting that, even though mortgage rates reached record lows in 2020 to 2021, the gap between mortgage rates and 10-year Treasury yields in fact reached peak levels not seen since the GFC—see [Figure 1](#).² However, the cause of this increase

¹From 2008 through 2019, annual mortgage originations never exceeded \$2.4 trillion, and they exceeded \$2 trillion only three times. Lending volume was \$3.7 trillion in 2003. Statistics are from *Inside Mortgage Finance* and can be seen on p. 8 of Urban Institute ([2024](#)).

²The effective duration of a 30-year mortgage is much shorter than 30 years due to prepayment.

in the mortgage-Treasury spread was different from the GFC (and from the recent 2022 to 2024 period when this spread also widened significantly): it is fully explained by a large rise in the “primary-secondary spread,” a measure of mortgage lenders’ markups.³ Another, conceptually sounder measure of markups, “gain-on-sale,” also increased sharply early in the pandemic and stayed elevated through to late 2021.

Figure 1

Intermediation markups typically rise during refinancing booms due to lender capacity constraints (Fuster et al., 2024). But this historical relationship accounts for less than half of the increase in 2020 to 2021. In other words, the elasticity of mortgage supply was abnormally *low*, despite the recent widespread adoption of online and digital technologies to streamline lending. We find evidence that operational and labor market frictions related to the pandemic can account for heightened capacity constraints in 2020 to 2021—e.g., licensing of new loan officers was disrupted, contributing to high labor utilization and wages for existing workers. Perhaps surprisingly, we also find little evidence that fintech lenders expanded supply more elastically than other nonbanks, further indicating that technology has not yet “solved” the problem of capacity constraints.

We also present novel evidence on the nature of capacity constraints in the mortgage market, concluding that these constraints today are effectively national in scope because the U.S. mortgage market is highly integrated geographically. Specifically we show that most mortgages are originated by lenders that operate in many regions, and that lenders,

Ten-year Treasury yields are therefore widely used as a benchmark for mortgage rates, even though this benchmark is imperfect, for reasons we explain below.

³We use the term “markup” throughout as shorthand for “gross markup” or “gross margin,” meaning in our case the difference between the marginal revenue of an intermediary and the marginal direct cost of funds lent to the borrower (but without accounting for wages, overhead, etc.), as in, e.g., Anderson et al. (2018).

and even individual loan officers, expanded to new markets in 2020 to 2021 to offset local differences in demand. As a result, there is little or no cross-sectional relationship between local demand shocks (or other local factors) and mortgage rates or origination timelines, even though these variables are tightly connected in the time series.

We then consider other potential drivers of high intermediation markups in 2020 to 2021. First, we study in detail the role of financial constraints facing nonbank lenders, a group often considered to be financially fragile. Studying the first major shock since the rise of nonbanks in the 2010s, we in fact find that nonbanks expanded lending *more* elastically than banks, aside from a brief period at the start of the pandemic. Further, even this initial drop in lending is more connected to lenders’ dependence on third-party originations (e.g., brokers) rather than capital or liquidity constraints. Nonbank lending fell more persistently in the jumbo market, however, reflecting the reliance of nonbanks on government-sponsored securitization.

Using variation across borrower types and locations, we are also able to rule out several other plausible explanations for high markups in the conventional conforming market (the main focus of our analysis⁴), including forbearance and default risk and the direct macroeconomic and health effects of the virus. High markups could also reflect a rise in lender market power, perhaps due to limited shopping by borrowers. We find little support for this channel, however: borrowers, if anything, searched more actively than usual during the pandemic, local market concentration decreased after the boom began,

⁴“Conventional” refers to mortgages not directly insured by the government (e.g., by the Federal Housing Administration or the Department of Veterans Affairs). Mortgages that are insured by the government-sponsored enterprises Fannie Mae and Freddie Mac are “conventional conforming” while conventional *nonconforming* loans include in particular “jumbo” loans with balances above the conforming loan limit.

and changes in mortgage rates were not closely connected to local concentration.

Finally, we extend our analysis beyond conventional conforming mortgages to explore credit supply dynamics in two other segments—the jumbo and Federal Housing Administration (FHA) markets. The jumbo market is informative because it lacks government guarantees and was not directly impacted by the Fed’s mortgage purchases under QE. This, and the fact that QE did not target “superconforming” mortgages for institutional reasons, allows us to isolate the distinct roles of guarantees and QE in supporting credit conditions, and we find that both played a role. However, we also find that these interventions were insufficient to fully stabilize the FHA market, where credit availability declined relative to the prime conforming market, particularly for the riskiest borrowers. We argue that these effects stem at least in part from the fact that FHA loans are riskier for lenders and servicers compared to conventional loans (Kim et al., [2018](#)).

We draw out several broader lessons from our results. (i) Despite recent technological advances, the U.S. mortgage market still faces capacity constraints in periods of peak demand, suggesting there are bottlenecks that technology has not yet been able to effectively address. (ii) Today, capacity constraints are primarily national in scope, because the U.S. mortgage market is geographically integrated, thus the key constraint on lending is *aggregate* industry resources. (iii) Nonbank mortgage lending may be more robust than previously appreciated, but remains dependent on government-sponsored securitization. (iv) Government guarantees support credit supply to riskier borrowers in the face of economic shocks, although, at least in the FHA market, these guarantees are not always sufficient to fully insulate borrowers. (v) QE has “local” effects on mortgage supply; even *within* the conforming market, QE particularly boosts supply in the segment where purchases are concentrated.

Our evidence contributes to a large body of research about the transmission of monetary policy and interest rate shocks through the mortgage market and the role of financial frictions, including Di Maggio et al. (2017), Berger et al. (2021) and Drechsler et al. (2024); see Amromin et al. (2020) for a review. Fuster et al. (2013, 2024), Sharpe and Sherlund (2016), Choi et al. (2022), and Frazier and Goodstein (2023) study how lender capacity constraints drive markups and lending during periods of high demand; we present new evidence that these constraints are national in scope and have not yet been alleviated through technology. We also contribute to research on mortgage QE (Krishnamurthy and Vissing-Jorgensen, 2011; Di Maggio et al., 2020) using novel variation in the intensity of Fed purchases within the conforming market, and provide new evidence on the effects of government guarantees on credit supply in times of stress, finding that guarantees support lending but are sometimes insufficient (see Calem et al., 2013; Vickery and Wright, 2013; Hurst et al., 2016, for related work studying earlier periods).

We also extend the literature studying the effects of post-GFC changes to the U.S. mortgage market—factors including tighter regulation and the rapid growth of nonbank and online lending (e.g., DeFusco et al., 2019; Gete and Reher, 2020; D’Acunto and Rossi, 2022). Research has identified several factors underlying the expansion of nonbanks and highlighted financial instability concerns associated with the nonbank business model (e.g., Buchak et al., 2018, 2024; Kim et al., 2018, 2022; Jiang, 2023). We find that nonbanks were resilient to the first major shock since their re-emergence in the 2010s, and in fact expanded lending more elastically than banks, except when liquid secondary markets were unavailable. We also present new evidence on the role of technology (see Buchak et al., 2018; Fuster et al., 2019; Jagtiani et al., 2021; Bartlett et al., 2022; Allen et al., 2023, for prior contributions), concluding that the rapid spread of online and digital lending has

not yet eased industry capacity constraints, at least based on the experience of the 2020 to 2021 boom.

Finally, we contribute to research about the mortgage market and consumer credit markets more generally during the COVID-19 pandemic (e.g., Bracke et al., 2020; Iverson et al., 2020; Cherry et al., 2021; Agarwal et al., 2024; Horvath et al., 2023) and to literature examining the role of technology for lending during this period (e.g., Ben-David et al., 2021; Kwan et al., 2021; Erel and Liebersohn, 2022; Branzoli et al., 2024).⁵

I. Data

In this section, we provide an overview of the different data sources we use in this paper.

Optimal Blue. Optimal Blue is a platform allowing lenders to access pricing information, initiate rate locks, and sell mortgages.⁶ Lenders are typically nonbank mortgage companies, but smaller banks and credit unions are also represented. Over 1,000 lenders and 200 investors use the platform, which is estimated to handle about one-third of U.S. mortgage originations in recent years. We use two forms of information produced by the platform.

Rate locks. These data reflect individual mortgage locks processed by Optimal Blue, covering around 280 metropolitan and rural areas. They include comprehensive under-

⁵Of course, the onset of COVID-19 led to severe disruptions in financial markets, and various spreads widened (see, e.g., Haddad et al., 2021; He et al., 2022). Siani (2024) shows that the gap in yields on corporate bonds between the primary market (at issuance) and secondary market trading also increased during the pandemic, by about 20bp, perhaps due to limited risk-bearing capacity by intermediaries given that the corporate bond market, like the jumbo market, does not feature government credit guarantees.

⁶Optimal Blue data (as referenced throughout) is anonymized mortgage market/rates data that does not contain lender or customer identities or complete rate sheets.

writing variables and offer several advantages over servicing data: e.g., they include data on discount points and credits, the exact rate lock date (as opposed to closing date, which happens much later), and the lock duration.

Mortgage offers. Optimal Blue’s “Pricing Insight” engine provides the real-time distribution of lender offers (combinations of rates and net points and fees) for a loan with given characteristics in a particular local market.⁷ This tool is designed for lenders to compare their pricing with that of their peers. Mortgage offer rates represent a direct measure of supply that can be observed regardless of whether the offer results in a loan in equilibrium. The Insight data also allow us to see how the number of lenders active in different segments of the market evolved over the pandemic.

Home Mortgage Disclosure Act (HMDA) Data. We use the confidential HMDA data, which covers nearly all U.S. mortgage applications (Bhutta et al., 2017). These data include a rich set of borrower and loan characteristics such as income, credit score, debt-to-income (DTI) ratio, loan-to-value (LTV) ratio, loan amount, property census tract, loan purpose (e.g., home purchase or refinance), and the exact application and closing date.

CSBS Nonbank Call Report and Consumer Access Data. We use quarterly data on the balance sheets of nonbank mortgage lenders from Mortgage Call Reports (MCR) to

⁷We have daily data in one local market (Los Angeles), twice-weekly data in four markets, and weekly data in 15 additional markets, collecting offer data for 100 loan types representing different combinations of FICO score, LTV ratio, loan program, purpose (purchase or cash-out refinance), occupancy, rate type (30-year fixed or 5/1 adjustable), and loan amount. For more details, see Bhutta et al. (2024), who compare mortgage offer rates and lock rates to study the efficiency of borrower search in the mortgage market.

study the relationship between lending and nonbank characteristics.⁸ To do this, we merge the MCR data with HMDA, matching on lender name. In total, we match 398 nonbanks accounting for 89% of total nonbank lending in the period from July 2019 to December 2020. We also obtain from the CSBS the Nationwide Multistate Licensing System (NMLS) Consumer Access data, which include the employment history of state-licensed and federally registered mortgage loan officers, which we use to study loan officer licensing activity.

Other Data Sources. For mortgage rates, in addition to Optimal Blue, we use data from the Freddie Mac Primary Mortgage Market Survey (PMMS) and the Mortgage Bankers Association (MBA). Mortgage-backed securities (MBS) pricing information comes from J.P. Morgan Markets. Mortgage servicing rights valuation data are provided by SitusAMC, an independent valuation service company. For direct evidence on lender income, costs, and employment, we use the MBA Quarterly Performance Report. Data on mortgage industry job postings for loan officers come from Burning Glass Technologies. Data on county-level daily COVID-19 cases come from the *New York Times* GitHub repository, and data on mobility come from Opportunity Insights. Metro area population data are from the 2018 5-Year American Community Survey, and unemployment data are from the Bureau of Labor Statistics’ Local Area Unemployment Statistics. Evidence on borrower search is obtained from Google Trends. Data on mortgage borrowers’ experiences in the application process come from the National Survey of Mortgage Originations (NSMO) released by the Federal Housing Finance Agency. Lastly, we use ICE McDash servicing data (“McDash data”) to study mortgage performance.

⁸These data are available to Federal Reserve researchers through an agreement with the Conference of State Bank Supervisors (CSBS), which owns and operates the system that collects MCR data on behalf of state regulators. Similar data are used by Jiang et al. (2020) to study the capital structure of nonbanks.

II. Rates and Markups for Conforming Mortgages

In this section, we decompose the mortgage-Treasury spread from [Figure 1](#) into different components and show that the increase in 2020 to 2021 was entirely due to a sharp rise in the primary-secondary spread, a measure of the intermediation markup in the primary mortgage market. We then present alternative measures of this markup—including our preferred one, gain-on-sale—and show that they paint a similar picture.

A. Decomposing the Mortgage-Treasury Spread

Define r_p as the mortgage rate paid by the borrower (in the “primary market”), r_{10} as the yield on a 10-year Treasury note, and r_s as the yield on a new-production MBS into which a typical newly originated mortgage with note rate r_p would be securitized (in the “secondary market”). We can decompose the mortgage-Treasury spread as:

$$r_p - r_{10} = \underbrace{r_p - r_s}_{\text{Primary-secondary spread}} + \underbrace{r_s - r_{10}}_{\text{MBS spread}}. \quad (1)$$

We can furthermore decompose the MBS spread as follows:

$$r_s - r_{10} \approx \underbrace{r_{\text{dur}} - r_{10}}_{\text{Duration Adjustment}} + \underbrace{\text{Option Cost}}_{\text{Option Cost}} + \underbrace{\text{Option-Adjusted Spread (OAS)}}_{\text{Option-Adjusted Spread (OAS)}}. \quad (2)$$

Duration Adjustment reflects the fact that the MBS may have a different duration from the 10-year Treasury. MBS duration is not known ex-ante, since it depends on prepayment behavior, so instead it is estimated using a model that simulates different interest rate and prepayment paths. *Option Cost* measures the value of the borrower’s prepayment option; the borrower can prepay at any time and will tend to do so especially when rates fall (to refinance into a new loan). Since MBS investors are short this option, they require compensation in terms of a higher MBS yield. Finally, *Option-Adjusted Spread (OAS)*

is a residual that captures various factors that affect relative pricing between Treasuries and MBS, such as liquidity, relative bond supply, perceived credit risk differences, and non-interest-rate prepayment risk (Boyarchenko et al., 2019).⁹

A.1. Measurement

While mortgage rates and Treasury yields are readily available, computing the decompositions in equations (1) and (2) is less straightforward. Most important, MBS yield, duration, option cost, and OAS are not directly observed, but are obtained from MBS pricing models; we rely on models from J.P. Morgan Markets, as noted earlier. Two further complications are that (i) MBS are traded in 50 basis point (bp) coupon increments, and a mortgage with a given note rate could be securitized into different new production coupons; and (ii) MBS investors do not receive the entire note rate since some cash flows are diverted to pay for the agency credit guarantee (g-fees) and servicing. To account for these factors we calculate a net note rate:

$$\text{Net Note Rate} = r_p - g - s. \quad (3)$$

For the guarantee fee g , we take the flow g-fee on new production MBS (as in Fuster et al., 2013); the (base) servicing fee s is set to 25bp, which is the market convention for MBS issued by the government-sponsored enterprises (GSEs) Fannie Mae and Freddie Mac. We then calculate the MBS yield, duration, option cost, and OAS by interpolating

⁹For textbook descriptions of MBS pricing, see Fabozzi (2016) or Davidson and Levin (2014). Note that the decomposition in (2) is not exact, since the OAS and the “zero-volatility spread” (the sum of OAS and option cost) are calculated as an average spread relative to each point on the interest rate curve based on which future cash flows are discounted, while the MBS spread is measured at a given point on that curve.

values between the two MBS coupons on either side of the net note rate.¹⁰

B. Decomposing the Mortgage-Treasury Spread over 2020 to 2021

Figure 2 traces out the evolution of the mortgage-Treasury spread in 2020 to 2021 (normalized to zero at the start of 2020) and decomposes this evolution into the four components discussed above. The mortgage-Treasury spread (the black dashed line in the figure) increased rapidly starting in late February, peaking at about 90bp above pre-pandemic levels in late March. The spread then gradually normalized, but even in August remained about 50bp higher than at the start of the year. The spread returned to pre-pandemic levels by November 2020, and thereafter stayed below its level at the start of 2020.

Figure 2

The decomposition shows that, except for a brief period at the onset of the pandemic, **about** the rise in the mortgage-Treasury spread in 2020 is *more* than entirely accounted for by **here]** a sharp increase in the primary-secondary spread—that is, by a higher markup in the primary mortgage market. In contrast, the three financial market components (duration adjustment, option cost, and OAS) were actually *lower* than their pre-pandemic levels over most of 2020 to 2021. OAS did spike temporarily in March 2020, reflecting an amplification of risk premia in financial markets and deleveraging by mortgage REITs.¹¹ Option cost

¹⁰Our method estimates the MBS yield for new mortgage production, and differs from the commonly used “current coupon” (CC) MBS yield, which is the hypothetical coupon trading at par. The CC yield is not reliable in 2020 since every coupon traded far above par. See Fuster et al. (2013) for related discussion.

¹¹While our measure of OAS is based on a pricing model from J.P. Morgan, [Figure IA.1](#) in the Internet Appendix shows that OAS evolved very similarly when using Citi’s pricing model. The Internet Appendix is available in the online version of the article on the *Journal of Finance* website.

also increased, reflecting higher interest rate volatility. But these spikes were short-lived. On March 15, the Federal Reserve resumed its MBS quantitative easing program, and on March 23 announced it would purchase agency MBS in the volume needed to support market functioning. These actions led to a rapid normalization of OAS, which then fell below pre-pandemic levels. The duration adjustment also became negative after February 2020, reflecting a drop in the model-implied duration of newly originated mortgages (from 4.7 years in January 2020 to 3.2 years by July).

For comparison, [Figure IA.2](#) in the Internet Appendix presents the same decomposition for two other episodes when the mortgage-Treasury spread was elevated: the 2007 to 2009 financial crisis period and 2022 to 2024 (featuring Fed rate hikes and “quantitative tightening”; see Drechsler et al., [2024](#)). The figure shows that the primary-secondary spread did not play a key role in either of these episodes. During the financial crisis, the high spread was mainly due to higher OAS reflecting elevated risk premia, while in 2022 to 2024 it was driven roughly equally by an increase in OAS, option cost (as rate volatility increased) and the duration adjustment, while the primary-secondary spread stayed flat.

Given these patterns, our focus will be on understanding the high primary-secondary spread, or more broadly high intermediation markups, during 2020 to 2021.

C. Alternative Measures of the Intermediation Markup

The primary-secondary spread is widely used to measure intermediation markups, but Fuster et al. ([2013](#)) and Fuster et al. ([2024](#)) argue that it is conceptually imperfect because it measures the instantaneous flow of income to intermediaries rather than the total gain reflecting the present value of these flows over the life of the mortgage. Due to prepayment, a mortgage has an uncertain and variable lifespan. Consequently, two

loans with the same primary-secondary spread may have very different expected incomes and values depending on anticipated prepayment behavior. A second challenge is that measuring MBS yield requires a pricing model; the measure is therefore model-dependent.

A preferred measure of the markup is provided by “gain-on-sale,” which takes the perspective of an intermediary selling the mortgage and the mortgage servicing right in the secondary market. Gain-on-sale is calculated as the present value of the net note rate (measured using secondary market MBS prices), plus the value of the servicing right (calculated using servicing multiples provided by SitusAMC), plus net points and fees. Details of our methodology are explained in Section [1.A](#) of the Internet Appendix, where we also present a more formal treatment of the gain-on-sale calculation.

The top two panels of [Figure 3](#) compare the primary-secondary spread and gain-on-sale over 2012 to 2022. The two evolve similarly, and roughly double between the start of 2020 and mid-April, with gain-on-sale peaking at just over 500bp. The two measures stay at extremely elevated levels through most of the rest of 2020 before normalizing over the course of 2021. (For more details, see [Figure IA.3](#) in the Internet Appendix, which zooms in on the 2019 to 2021 period and also shows that these patterns are similar if we use alternative series for the primary mortgage rate.) By either measure, intermediation markups were much higher in 2020 to 2021 than any other point in the sample period. **Figure 3**

We can use gain-on-sale directly to get a sense of just how profitable mortgage lending **about** was during this period. Focusing just on the last three quarters of 2020, originations averaged \$1.2 trillion a quarter; multiplying originations by our quarterly gain-on-sale measure **here]** implies total industry gain-on-sale of \$171 billion, or \$57 billion a quarter, compared to a quarterly average of only \$12 billion in 2019. This huge increase reflects the product of higher lending volume and higher margins. But even holding originations fixed, had

gain-on-sale remained at its pre-pandemic level of 2.5%, the industry would have earned only \$88 billion from 2020:Q2 to 2020:Q4—about half as much as it did.

The bottom two panels of [Figure 3](#) display measures of mortgage origination profit margins derived from SEC filings and MBA survey data. These measures have their limitations—e.g., they are average rather than marginal—but they tell a similar story, and suggest that mortgage lending was more profitable in 2020 to 2021 than ever before or since.¹²

III. Capacity Constraints

In this section, we present evidence that capacity constraints amplified by pandemic-era operational and labor market frictions provide a coherent explanation for the strikingly high intermediation markups observed in 2020 to 2021.

We start by showing that high markups are partially but not fully explained by the high level of mortgage demand—in other words, credit supply was unusually inelastic compared to prior booms. We find that operational and labor market issues (e.g., disruptions to loan officer licensing) made it more difficult for lenders to expand their workforces and led to longer processing times and closing delays, exacerbating “usual” capacity constraints documented in Fuster et al. ([2024](#)). Further, we present novel evidence that these capacity constraints were national in scope because the mortgage market is geographically

¹²For example, Rocket Companies, the largest U.S. mortgage lender, recorded \$9.4bn in net income in 2020, up over 900% from 2019. More broadly, if we multiply net production income per dollar of originations by average origination volume per firm in the MBA data, the resulting total net production income skyrocketed from below \$5mn in 2020:Q1 to an average of \$21.4mn over Q2-Q4, with a peak of over \$27mn in Q3.

integrated. Finally, we study whether digital and online lending has ameliorated capacity constraints—here, 2020 to 2021 is a key litmus test since it was the first refinancing boom since the widespread adoption of these technologies.¹³ We in fact find little evidence that technology-based lenders expanded lending more elastically than other nonbanks; this and the low overall elasticity in 2020 to 2021 suggest that technology has not yet “solved” the problem of capacity constraints.

A. Was 2020 to 2021 Consistent With Historical Patterns?

We first study whether the historical relationship between intermediation markups and capacity utilization can account for the high markups in 2020 to 2021. [Figure 4](#) plots mortgage demand against the two measures of markups we have used already: the primary-secondary spread and the gain-on-sale.¹⁴ Our preferred proxy for demand is the difference between the weighted average coupon (WAC) on the stock of mortgages and the 10-year Treasury yield. This measure is highly correlated with applications for (refinance) loans and is arguably exogenous to mortgage supply shocks—or more precisely, the supply of intermediation in the mortgage market—because it does not rely on current mortgage

¹³Online lending was rare up to the mid-2010s (Buchak et al., [2018](#); Fuster et al., [2019](#)), but by 2020, 91% of lenders offered digital applications through an online portal (ICE Mortgage Technology, [2021](#)).

¹⁴Note that for scatter plots and regressions, we adjust the Freddie Mac PMMS rate so that it reflects the rate for a loan with zero points. To do this, we take the mortgage rate and add the average points reported by Freddie Mac multiplied by a weekly point-rate trade-off. We estimate this point-rate trade-off using Optimal Blue Insight data, where we observe the offered interest rates for loans with different numbers of net points (+2, +1, 0, -1, and -2). For years prior to 2017 when Optimal Blue data are not available, we estimate this point-rate trade-off by using MBS prices at different coupons (which gives similar results).

rates.¹⁵ We alternatively use the MBA mortgage applications index as a direct measure of demand. (These two series typically co-move closely; see Panel C of [Figure 1](#).)

Figure 4

[Figure 4](#) shows that markups during the pandemic significantly exceed what would be predicted simply from the level of demand. Red squares in the figure are from 2020 and green diamonds from 2021, with the number indicating the month. Blue circles are from 2012 to 2019. There is a clear positive historical relationship between markups and mortgage demand. But from about April 2020 onward, markups are well above the line of best fit, before declining later in 2021, regardless of which demand or markup measure is used.

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[Table I](#) quantifies the “excess” markup by regressing the primary-secondary spread or gain-on-sale on either demand proxy as well as time dummies corresponding to different phases of the pandemic. In all cases, the pandemic dummies indicate large, statistically significant excess markups from March 2020 through the first half of 2021. The historical relation between markups and demand can account for only about 20% to 45% of the rise in intermediation markups, depending on the specification.^{16,17}

Table I

¹⁵In contrast, using the spread between WAC and the current *mortgage* rate may produce bias in the estimated markup relationship, because mortgage rates reflect the intersection of demand and supply. Note that we study the equilibrium price in the primary market for mortgage *intermediation*; thus the requirement for our demand proxy is that it does not affect the supply of intermediation services, which appears reasonable for the WAC–10-year Treasury spread. See Fuster et al. (2024) for more detailed discussion.

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¹⁶On the high end of this range, the coefficient in column (4) would project a maximum increase in gain-on-sale of 123bp, given that the MBA applications index increased from an average of 533 in January 2020 to a maximum of 1,204. This compares with an actual increase in gain-on-sale of 273bp.

¹⁷The pandemic dummies from March through September are comparatively smaller for gain-on-sale than for the primary-secondary spread, taking into account that the latter is a flow measure while the

Specifically, column (1) shows that the primary-secondary spread is 73bp higher than expected in March and April 2020, and 81bp higher than expected in May through September. For comparison, the one-standard-deviation (std) residual from this regression excluding 2020 to 2021 is only 8.9bp (last row of table). Thus, one could consider the COVID-19 period roughly a “9 sigma” event for over six months before its effects slowly subside through to late 2021. Results are similar in column (2). The excess markup is also highly significant when we use gain-on-sale as the dependent variable (columns (3) and (4)). Gain-on-sale is \$0.91 to \$1.13 higher than expected in March-April 2020, rising to \$1.21 to \$1.59 in May-September 2020, and remains highly elevated until the second half of 2021.

B. Operational Constraints in Expanding Capacity

Next, we present evidence that operational and labor market issues related to the pandemic made it more difficult than usual for lenders to expand capacity and can account for the low elasticity of mortgage credit supply in 2020 to 2021.

Much of our evidence focuses on labor market frictions in hiring employees to cope with the surge in demand. Our analysis is motivated by anecdotal reports from mortgage industry participants that the abrupt, unexpected shift to a remote-work environment created significant challenges in hiring and training loan officers, processors, and other workers.¹⁸ Labor supply was further held back by pandemic-related disruptions to the former measures the total gain earned by intermediaries. This reflects (i) a flattening of the relationship between gain-on-sale and the primary-secondary spread during the pandemic, and (ii) the fact that gain-on-sale is below the line of best fit at the start of 2020, while the primary-secondary spread is close to its fitted value.

¹⁸E.g., one practitioner told us it was difficult to remotely train and monitor new employees; as a result

licensing of mortgage loan officers through the NMLS, a process involving background checks, fingerprinting, an exam, and ongoing education.¹⁹ Early in the pandemic, half of fingerprinting locations were shuttered, with 10% still closed in December 2020. Testing sites also closed initially, although remote testing became available in September.²⁰ The pandemic also created operational challenges in closing loans—e.g., office closures made it difficult to document borrower employment and income, high layoffs often required checking employment multiple times (Berry and Kline, 2020), and county recorder offices were closed or on limited schedules (Hughes, 2020).

We now present evidence on the effects of these operational and labor market frictions on the quantity and cost of labor, loan officer licensing, and delays in loan closings.

B.1. The Quantity and Cost of Labor

One test of whether lenders had difficulty in expanding employment is whether there was an unusual increase in the capacity utilization of the existing labor force. Evidence on

lenders prioritized hiring experienced, trusted professionals (generally from competitors) requiring less training and oversight. Similarly, an executive at nonbank Mr. Cooper explained their limited growth as follows: *“It reflects the fact that we add capacity at a deliberate pace with an eye on the long term. And additionally, when the crisis hit and we shifted to work-from-home status, that slowed our hiring and our onboarding”* (Ivey, 2020).

¹⁹A new license is required for de novo loan officers as well as officers shifting from banks to nonbanks or moving across states. In these latter cases, the officer may temporarily operate for 120 days before obtaining a new license (Mortgage Bankers Association, 2019). We were told, however, that in some cases licensing delays exceeded this 120-day grace period.

²⁰One practitioner told us by email that: *“Testing centers closed at times and have had limited availability. This has prevented new LOs [loan officers] from joining the industry right away. We have seen delays of anywhere from a few weeks to a few months depending on the restrictions in the area.”*

this point is presented in Panel A of [Figure 5](#), which plots the number of loans originated per sales employee (i.e., loan officer), using MBA data. Monthly originations per employee jumped from 5.3 in 2020:Q1 to 7.5 in Q2, and then exceeded 8 until 2021:Q2.²¹ Labor utilization usually rises in periods of high demand, but as the figure shows, this does not account for the levels reached in 2020 to 2021, which also far exceed any point over the previous decade including the 2012 refinance boom.²²

Figure 5

MBA data also show that the price of labor was unusually high (Panel B of [Figure 5](#)). **about** Labor costs per dollar of lending typically fall significantly during booms due to scale **here]** economies. But during the pandemic, sales employee costs scaled by loan volume were 20 to 25 bp higher than expected based on historical patterns, equivalent to more than 10% of average total personnel costs measured over the previous five years.²³

One possibility is that lenders did not want to hire more in 2020 because they thought the boom would be short-lived. This is not consistent, however, with Panel C of [Figure 5](#), based on Burning Glass data, which shows a dramatic increase in new job postings for loan officer positions early in the pandemic. Monthly postings peaked in August 2020; overall, in 2020:Q3, the volume of job postings was more than 2.5 times its level one year

²¹Originations in the first quarter of 2020 were still relatively low because they mostly reflect applications from the last quarter of 2019, before the surge in refinancing activity.

²²[Figure IA.7](#) in the Internet Appendix plots the time series of originations per sales employee and per employee overall. Although labor utilization for nonsales employees also increased in 2020, this uptick aligned with patterns observed in earlier periods of high demand (e.g., 2012), suggesting that the labor shortage in 2020 to 2021 was most acute for loan officers and other sales personnel.

²³Notably, personnel costs stayed high in 2021 even as capacity constraints started to ease, likely because loan officers were able to lock in attractive compensation packages at the peak. We show in [Figure IA.7](#) in the Internet Appendix that compensation costs for nonsales employees were also unusually elevated, although to a lesser degree in line with the evidence on quantities discussed above.

earlier. The overall level of postings in 2020 to 2021 far exceeded any other period since 2010.

B.2. Loan Officer Licensing

Next, we test whether loan officer licensing volume was unusually low by comparing realized outcomes to a counterfactual computed as the out-of-sample prediction from a time-series regression of $\log(\text{licenses})$ on four lags of $\log(\text{mortgage applications})$ and a December seasonal dummy. We construct licensing volume by aggregating NMLS micro-data. The model is estimated using pre-pandemic data from February 2015 to February 2020.²⁴

Results reported in Panel D of [Figure 5](#) indicate a significant licensing deficit. The model predicts a rise in licensing in 2020 to 2021 reflecting the growth in applications. But instead, license issuance dropped by about half in spring 2020 and even the flow rate did not recover to its pre-pandemic level until late 2020. Integrating the difference between “actual” and “counterfactual” licensing implies a cumulative deficit in licenses issued of 11,200 by October 2020.²⁵

²⁴See [Table IA.I](#) in the Internet Appendix for the regression coefficients. The counterfactual series plotted in [Figure 5](#) is based on our preferred specification, column (1). Licenses typically track loan applications with a two-month lag. We include only a dummy for December, not the other calendar months, because of the clear year-end drop in licensing evident in Panel D of [Figure 5](#).

²⁵Over the previous five years, the average number of new licenses over the March-October period was 46,700, meaning the deficit amounts to about one-quarter of that number.

B.3. Processing Times and Delays in Loan Closing

Finally, we study the effects of labor shortages and other operational bottlenecks on mortgage processing. We first focus on processing times as measured in the confidential-use HMDA data, defined as the gap between application and origination dates (following, e.g., Fuster et al., 2019, 2024). We regress processing time in days (winsorized at the 1st and 99th percentile) on borrower and loan characteristics (including loan purpose), county dummies, and dummies for loan application month. Panel A of **Figure 6** traces out the coefficients on these month dummies, with February 2020 as the base category. The figure shows that processing times were substantially higher from March 2020 onward, initially by about 6 days, rising to 11 to 13 days at the peak from July to September 2020. This is a large increase relative to the pre-pandemic average processing time of about 48 days. (**Figure IA.8** in the Internet Appendix plots the evolution of median processing time over 2012 to 2021 and shows that it moves closely with application volume, especially for refinances.)

Figure 6

Second, we use NSMO data to directly study closing delays. This survey asks borrowers **about** four yes/no questions related to delays in the mortgage closing process. We construct **here]** a dummy equal to 1 if a respondent answered yes to any question.²⁶ Similar to the processing time analysis, we regress this dummy on origination month dummies and loan controls (e.g., credit score, loan size bins, income, components of wealth, risk aversion, race, ethnicity). Panel B of **Figure 6** traces out the monthly coefficients, showing a

²⁶Borrowers are asked whether: (i) they had to redo/refile paperwork because of loan processing delays; (ii) they had to delay or postpone their closing date; (iii) loan documents were not ready at closing; (iv) the closing did not occur as scheduled. The pattern of our results is very similar if we use a composite indicator summing across the four questions (with values ranging from 0 to 4) instead of a dummy.

significant increase in the propensity to experience a delay in loan closing from August 2020 onward.²⁷ The magnitude of the effect is quite large—the estimated incidence of any delay is 30% to 50% higher in the second half of 2020 than in the pre-pandemic period.

C. Are Capacity Constraints Local or National?

While we have so far focused on mortgage rates at the national level, it is quite striking that rates evolved very similarly across the country, as illustrated in **Figure 7**. Panel A shows (based on Optimal Blue Insight data) that although mortgage rates do vary across metro areas, the differences were consistently small over 2020 to 2021. Panel B, looking in the cross-section, shows that mortgage rates fell by similar amounts across metros during the pandemic (comparing March-December 2020 to October 2019-February 2020), generally within a few basis points, as seen by the proximity of most metros to the 45 degree line.

Figure 7

What explains this uniformity, and how can it be reconciled with a capacity constraints **about** interpretation of the evidence, given that there was local variation in mortgage demand, **here]** economic conditions and the spread of the virus? We argue there are two main parts to the story. First, as in prior refinancing booms, the surge in mortgage demand was in fact quite aggregate in nature (Panel C of **Figure 7**). While there is of course some cross-market variation, the figure shows that demand increased strongly everywhere as rates fell; e.g., year-over-year application growth averaging over 2020 was 42% at the 25th percentile of the distribution compared to 58% at the 75th percentile.

²⁷This timing broadly lines up with Panel A, since in Panel B time is indexed by origination month, which is typically about two months after the application month but likely longer for loans experiencing delays.

Second, the U.S. mortgage market is, to an underappreciated degree, highly integrated geographically, with human resources and capital able to shift quite freely across locations in response to local shocks.²⁸ Specifically, we find novel evidence that lenders, and even individual loan officers, were geographically diversified even prior to COVID, and further that this geographic diversification *increased* as demand surged in 2020, despite pandemic-era frictions that could conceivably have constrained resource flows.

Evidence on these points is presented in Panel D of [Figure 7](#), which constructs measures of the geographic dispersion of lending using HMDA data. For each application we compute the number of distinct core-based statistical areas (CBSAs, effectively a metro area) where the lender, and also the loan officer of record, was active in the same month, where “active” means they received at least one application. We then take the mean and median of these “active CBSAs” variables across all HMDA applications in the month.

We find that pre-COVID, the originator of a typical loan was active in 250 to 350 CBSAs in the same month, indicating a high degree of geographic dispersion.²⁹ While there is a long tail of small lenders, most loans are made by intermediaries active in

²⁸As concrete examples, the two largest lenders in our sample period, Rocket Companies and United Wholesale Mortgage (UWM), operate in all 50 states without any significant local labor market presence at all. Instead, the workforces of both these firms are centered in their head offices in Michigan, to be shifted across markets or products as needed. (E.g., the 2022 10-K for UWM states: “*As of December 31, 2022, we had approximately 6,000 team members, substantially all of whom are based in our corporate campus in Pontiac, Michigan.*”). As further illustration of geographic integration, it is standard for lenders to centralize mortgage underwriting and processing through regional or national hubs. E.g., for Wells Fargo, many back-office activities are centralized in Des Moines, Iowa, and a second mortgage hub near Minneapolis (Weil, [2019](#)).

²⁹For reference, there are 849 CBSAs in the data, of which only 353 have a population of 100,000 or more.

many markets able to shift financial capital and organizational resources across regions depending on demand. Moreover, lenders expanded to new markets in 2020 to 2021 to become progressively *more* geographically diversified, aside from a temporary blip in March 2020.

Even if lenders are national, local capacity constraints might still bind if individual mortgage professionals are closely tied to individual markets. Panel D of [Figure 7](#) also finds, however, that in 2019 the loan officer for a typical mortgage operated in a mean of nine different CBSAs in the same month, and a median of three CBSAs. Furthermore, as demand spiked in 2020, loan officers significantly expanded the number of markets in which they operated—the average increased from about nine to 12, while the median rose from three to five.

To confirm that these facts are not due to compositional shifts, [Table II](#) estimates linear models at the lender or loan officer by month level tracing out how “active CBSAs” (measured in logs) changes over time conditional on lender or loan officer fixed effects. Columns (1) and (4) show that geographic dispersion of lending activity is strongly positively related to the volume of applications received. Columns (2) and (5) show that dispersion increased after March 2020 (relative to the prior six months), by up to 30% at the loan-officer level and 20% at the lender level. For lenders, the rise in the number of active CBSAs is explained by the high rate of application growth; for loan officers, geographic dispersion rose even more than application growth would predict (columns (3) and (6)).

Table II

The ability of lenders and loan officers to shift across locations and enter new markets **about** as needed implies that the key constraint on credit supply is *aggregate* industry resources **here]** rather than local capacity tied to the specific market. As a further test of this hypothesis,

we examine directly whether capacity constraints (measured by processing time or mortgage rates) in 2020 were more binding in locations experiencing faster application growth. Evidence is reported in the form of binned scatter plots in [Figure IA.9](#) in the Internet Appendix. Panels A and B of the figure show that the large increase in processing time we documented earlier is remarkably insensitive to local year-over-year application growth, while Panel C finds the same thing for mortgage interest rates.

In short, while we have seen earlier that mortgage demand correlates strongly with intermediation markups and processing times in the time series (see also Fuster et al., [2024](#)), there is little or no relation in the cross-section, reflecting that industry resources are sufficiently mobile to offset local demand shocks.

D. Did Technology Help?

The low mortgage supply elasticity in 2020 to 2021 is particularly striking in light of the recent wave of technological innovation to digitize lending and put the application process online (Buchak et al., [2018](#); Fuster et al., [2019](#)). While such technology has significant promise to mitigate capacity constraints through automation, the fact that credit supply was *less* elastic in 2020 to 2021 than prior booms suggests, *prima facie*, that recent technological change has not significantly relaxed industry capacity constraints.

That said, it is possible that new technology did help, but its benefits were overwhelmed by the operational and labor market frictions discussed earlier. Therefore, as a more direct test, we study the performance of technology-based, or “fintech,” lenders, specifically whether such lenders were able to increase lending more rapidly or process loan applications faster than other originators during the 2020 to 2021 boom.

We use a classification of fintech lenders from Jagtiani et al. ([2021](#)), which itself builds

on Buchak et al. (2018) and Fuster et al. (2019).³⁰ These classifications are based on whether lenders offered online applications as of 2016, but we posit that it may still be reasonable to rely on them in 2020 to 2021 if these “early adopters” (e.g., Rocket Mortgage) remain near the technological frontier. Supporting this assumption, Figure IA.10 in the Internet Appendix shows using HMDA data that fintechs were indeed still able to process observably similar loans 6 to 7 days faster than other nonbanks in late 2019 and early 2020, just before the pandemic.³¹ (Interestingly though, we find the “fintech advantage” in processing speed has diminished over time as online lending has become widespread, by about half a day per year.)

We then estimate linear models using loan-level HMDA data in which we regress a dummy for whether the lender for a given loan is a fintech on a pandemic dummy and loan characteristics, thereby measuring the (conditional) change in the fintech market share. We restrict the sample to conventional conforming originations with an application date between July 2019 and December 2020; that is, we focus on the first nine months of the boom when markups were particularly high and a “pre” period of similar length.³²

³⁰The lenders classified as fintechs are: AmeriSave, Better Mortgage, CashCall, Homeward, Everett Financial (Supreme), Guaranteed Rate, LoanDepot, Movement Mortgage, SoFi, and Quicken (Rocket Mortgage).

³¹Results are based on a loan-level linear model of processing time estimated month-by-month from January 2012 to December 2021, conditioning on the controls from Table 3 of Fuster et al. (2019). We estimate results separately for purchase loans and refinances, restricting to conventional conforming first-lien mortgages.

³²We use HMDA data through 2021 and consider all loans with an application date within our sample window even if the loan did not close until 2021. This is important to avoid selectively excluding some loans that were slow to close. Table III includes both purchase mortgages and refinances, although we estimate results separately by loan purpose in Table IA.II in the Internet Appendix; results are generally

Results are presented in [Table III](#). In column (1), which includes banks and nonbanks and has no controls, we find a significant 2.6 percentage point (pp) increase in the fintech market share during the pandemic, compared to a sample average of 16%. However, the rise in the fintech share is significantly smaller when we restrict the sample to nonbanks (column (2)), a more comparable sample because all fintechs are nonbanks and the non-bank share of lending rose in 2020 to 2021 (see [Section IV](#)). And after also controlling for loan characteristics (column (3)), the change in the fintech share is essentially zero—the confidence interval rules out an increase larger than 0.5pp.³³ Column (4) of [Table III](#) does, however, find that fintechs pivoted towards low-credit-score loans, which Sharpe and Sherlund ([2016](#)) argue are rationed during booms because they are more labor-intensive to underwrite, suggesting technology may have promoted the flow of credit in that segment.³⁴ [Table III](#)

Columns (5) to (7) instead study processing time in days. The negative coefficient on [about here\]](#)

³³While we include a rich set of controls (see notes to [Table III](#) for the full list), the change in the coefficient from columns (2) to (3) is similar if we control only for a refinancing dummy. This reflects the fact that fintechs specialize in refinances, which became a larger overall share of lending during the pandemic. Looking beyond average effects, [Figure IA.11](#) in the Internet Appendix plots the dynamics of the fintech share of nonbank conforming lending separately for purchase mortgages and refinances, based on the regression model from column (3) of [Table III](#) but replacing the pandemic dummy with month dummies. For purchase loans, the fintech share rises at the onset of the pandemic, by up to 2pp, but the effect dissipates by June. For refinances, the fintech share is quite volatile from month to month, meaning that estimates of how the fintech market share changed during the pandemic depend on the length of the pre- and post-period chosen. Finally, we note that in fact this evidence is quite consistent with what Fuster et al. ([2019](#)) find for an earlier sample period: while fintech lenders were able to process loans faster when application volumes rose, they did not expand market share.

³⁴Consistent with Sharpe and Sherlund ([2016](#)) and Frazier and Goodstein ([2023](#)), [Appendix B.3](#) shows that processing times on low-FICO mortgages are indeed higher in the conventional conforming market.

“fintech” confirms that fintechs were able to process mortgages faster than other lenders prior to the boom. While fintechs were also faster during the pandemic itself, the small positive interaction term shows that the “fintech advantage” over other nonbanks if anything *declined* slightly during the boom, by up to two days.³⁵ This again suggests that capacity constraints during the pandemic were at least as binding for fintechs.

In short, while revealed preference shows consumers value the speed and convenience of online lending, evidence from the pandemic suggests that technology has not yet “solved” the problem of capacity constraints during periods of peak demand, perhaps due to bottlenecks in the origination process that are hard to remove by technology alone.

IV. Alternative Mechanisms

We now consider several other plausible drivers of high markups in 2020 to 2021. First, we study the role of financial constraints faced by nonbank lenders as a potential brake on credit supply. We find that the nonbank share of lending indeed fell temporarily early in the pandemic, although this seems related to nonbanks’ greater reliance on third-party brokers rather than capital or liquidity constraints—further, after this initial dip, nonbanks expanded lending *more* than banks. We then turn to explanations related to

³⁵In contrast, Fuster et al. (2019) find during an earlier period that processing speed is less sensitive to credit demand shocks for fintechs than other lenders. The difference between our findings and these earlier results may reflect the widespread diffusion of online lending over the intervening period. Note also that the monthly coefficients shown in Appendix Figure IA.10 indicate that for refinances, fintech lenders’ processing speed advantage over other lenders fell sharply over the first four months of the pandemic, but then increased again and was significant during the remainder of 2020 to 2021. For purchase loans, initial relative changes were small but then fintech lenders became faster in the second half of 2020, especially Q4.

default and forbearance risk, direct health and economic effects of the virus, and lender market power, finding that none of these factors played an important role.

A. Nonbank Financial Frictions

Our analysis of lender financial constraints focuses on nonbank mortgage companies, which faced significant liquidity strains early in the pandemic, including margin calls on hedges, liquidity outflows due to mortgage forbearance, and concerns about the continued availability of warehouse funding (Loewenstein, 2021; Pence, 2022). These stresses are important because nonbanks are now central to the U.S. mortgage market, having originated the majority of loans in recent years (Buchak et al., 2018; Kim et al., 2022). Nonbanks are fragile because they are financed by short-term wholesale debt and lack access to liquidity backstops available to banks such as the discount window and Federal Home Loan Bank advances (Kim et al., 2018; Jiang et al., 2020). Banks also experienced large deposit inflows in 2020, and Li et al. (2020) find they were not liquidity constrained as a result.

We test whether financial frictions constrained nonbank lending in two ways. First, we examine whether, in aggregate, the nonbank share of lending fell as the pandemic took hold. Second, we study the cross-section of nonbanks to see whether mortgage companies with low liquidity or capital ratios lost market share to rivals with stronger balance sheets. Our analysis is based on HMDA data merged with nonbank characteristics from the Mortgage Call Reports, as described in Section I.

A.1. Time-Series Shifts in the Nonbank Share

Figure 8 plots the evolution of the nonbank share of conforming mortgage lending, as well as the conditional share controlling for county fixed effects and/or loan characteristics. Time is indexed by application date, the more relevant date for loan pricing and credit availability. The nonbank share initially increased in February and early March 2020 as mortgage applications spiked, but then fell sharply starting in mid-March—the unconditional nonbank share dropped by about 5pp while the conditional share fell by about 3pp. This decline was short-lived, however. The nonbank share surpassed its pre-pandemic level by late April, then rose further, reaching 64% by December 2020, a full 9pp *higher* than a year earlier. These patterns are robust to the controls used.

Figure 8

This suggests financial frictions may have constrained nonbank lending, but if so, the effect lasted only a month or two. This seems consistent with the chronology detailed in Loewenstein (2021) and Pence (2022). Nonbanks faced significant financial pressure in March and April, sparking calls for an industry bailout; but by May, nonbank liquidity began to benefit from the prepayment “float” generated by the refinancing wave.³⁶ Government actions also stabilized the industry: e.g., Ginnie Mae announced a temporary backstop facility for nonbanks on April 10 (Ginnie Mae, 2020), and on April 21, the FHFA set a four-month cap on servicing advances for loans in forbearance (FHFA, 2020).

Extending this graphical evidence, **Table IV** reports estimates from linear models in which we regress a dummy for whether the lender is a nonbank on time dummies and

³⁶Cash inflows from prepayments are retained by the servicer temporarily before the funds are forwarded to investors (see Pence 2022 for details). Although mortgage application volume started rising sharply in late February, it took a couple of months before this translated into a rising “float” for servicers, because during this period it took about 60 days on average for a typical loan to be closed and funded.

various controls.³⁷ Column (1) traces out the raw nonbank share of conforming lending, which rises by 0.8pp early in the pandemic, drops to 1.3pp *below* pre-pandemic levels during the peak stress period (March 13-April 30), then rises to 6.8pp above pre-pandemic levels on average in May-December. Results with controls are similar (columns (2) and (3)).

Table IV

Columns (4) and (5) focus on low-credit-quality loans that may have more funding risk **about** (e.g., because of a higher risk of defects that disqualify loans from being securitized). The **here]** nonbank share does indeed rise by less for loans with low credit scores and high LTVs, but nonbanks still gain significant market share for such loans. Finally, column (6) drops fintechs from the sample to check whether the rise in the nonbank share simply reflects a shift to online lending; results show that this is not the case, consistent with Section **III.D**.

A.2. Cross-Sectional Evidence

We turn to cross-sectional analysis to shed light on the mechanisms underlying these shifts. Financial constraints are a natural explanation for the temporary dip in the nonbank share in March-April 2020, but other stories are also plausible. For example, nonbanks rely more heavily on brokers and correspondent lending—so-called “third-party originations” (TPOs)—and the third-party channel declined sharply at the start of the pandemic, likely because economic uncertainty led lenders and investors to want tighter control and oversight over the origination process (Recursion Inc. 2022; Figure IA.12 in the Internet Appendix).

To proceed, we collapse the HMDA-MCR data by lender and week, and estimate a

³⁷The “base” pre-period is December 2019 to mid-February 2020; we use a short pre-period because of the upward drift in the nonbank share earlier in 2019 evident in Figure 8.

Poisson model of conforming lending as a function of nonbank characteristics (measured as of 2019:Q4) during different phases of 2020. Results are reported in [Table V](#).³⁸ We standardize each characteristic so that coefficients measure the percentage change in lending for a one-standard-deviation change in the characteristic. Specifications include lender and week fixed effects, and standard errors are clustered by lender.

Table V

We consider two measures of financial condition: liquidity as measured by the ratio of cash to total assets, and the equity capital ratio. In both cases, we find little indication that nonbanks with stronger balance sheets expanded lending faster, either in the univariate specifications in columns (1) to (5) or the multivariate regressions in columns (6) and (7). This is true even during the “peak stress” period (mid-March to April 2020), where the point estimates on capital and liquidity are indeed positive, but not significant.

We do however find evidence that mortgage companies reliant on a TPO business model experienced lower lending growth, at least initially.³⁹ In column (6), a one-standard-deviation increase in the ex-ante TPO share is associated with 8% lower lending in the “early pandemic” period and 18% lower lending in the “peak stress” period, significant at the 1% and 10% levels, respectively. The effect dissipates over the remainder of 2020,

³⁸We study the full cross-section of nonbanks, but since nonbank lending is quite concentrated, column (7) of [Table V](#) re-estimates the multivariate model from column (6) restricting the sample to nonbanks with at least \$1bn in assets as of 2019:Q4. The resulting point estimates are quite similar to column (6), although the estimates are less precise due to the much smaller sample (this screen reduces the sample size by 92%).

³⁹We measure reliance on third-party originations using a HMDA variable for whether the borrower applied for their loan through a third party rather than directly to the lender. We define “TPO share” as the mean of this variable at the lender level, measured over the five months prior to our regression sample period.

although the point estimate remains negative.⁴⁰

Interestingly, we also find that smaller nonbanks expanded lending faster in 2020 (statistically significant except for the “peak stress” period), perhaps because small organizations were more nimble in a rapidly changing environment. This result is further evidence that financial constraints were not a major drag on lending, since such constraints would likely be amplified for small nonbanks. Finally, the model includes a fintech dummy; consistent with Section III.D, we find little evidence of a significant shift in the fintech share of nonbank lending over 2020, with or without other lender controls.

Looking outside the conforming market, Figure IA.13 in the Internet Appendix finds a larger and more persistent decline in the nonbank share of lending in the jumbo market. Why? Nonbank jumbos are typically sold to banks, who likely sought more control over underwriting given that jumbos are not guaranteed;⁴¹ the pandemic also saw a drop in jumbo securitization, the other main funding option for nonbanks (Federal Reserve Board, 2021). In contrast, the nonbank share of FHA/VA lending rose in 2020 to 2021 similarly to the conforming market (Figure IA.14 in the Internet Appendix), reflecting continued funding through the government-backed agency MBS market. These results highlight the dependence of nonbank mortgage lending on the availability of liquid secondary markets

⁴⁰We also re-estimated our loan-level model including the TPO dummy as an additional control as a different way to evaluate this channel. Results shown in Table IA.III in the Internet Appendix are consistent with Table V: controlling for loan channel increases the coefficient on the nonbank dummy in the first two pandemic stages and somewhat reduces the drop in the nonbank share between the “early pandemic” and “nonbank stress” periods.

⁴¹E.g., Wells Fargo entirely stopped purchasing jumbo mortgages from third parties early in the pandemic, even though it continued to originate some jumbos through its own retail channel (Eisen, 2020).

(as also studied in Buchak et al., 2024).

B. Forbearance and Default Risk

The pandemic led to a surge in mortgage forbearance and nonpayment, reflecting high unemployment and a CARES Act provision requiring servicers to provide up to a year of forbearance on federally backed mortgages.⁴² Nonpayment is costly for mortgage intermediaries, even in the conforming market where loans are securitized and carry a credit guarantee. First, there is liquidity risk because the servicer must advance mortgage payments, taxes, and insurance even if the borrower stops paying. These advances will eventually be reimbursed, but funding them in the interim may be expensive or infeasible.⁴³ Second, there is “pipeline” risk that the loan becomes delinquent before it can be sold.⁴⁴ Third, servicing a delinquent loan is much more labor intensive and costly.

We use cross-sectional variation to study whether this heightened default risk led to a risk premium that increased mortgage rates for conforming borrowers. Our approach uses the fact that the increase in the nonpayment rate was up to an order of magnitude higher for low-credit-score borrowers (see Section VI.A of the Internet Appendix). If

⁴²At the peak in June 2020, 8.6% of mortgages were in forbearance (source: MBA). The total mortgage nonpayment rate more than doubled from 3.2% in January 2020 to 7.8% by May (Black Knight, 2020).

⁴³This was a cause of great concern in the industry early in the pandemic. On April 21, 2020, it was announced that servicing advances for conforming mortgages securitized by Fannie Mae and Freddie Mac would be capped at four months of principal and interest (FHFA, 2020). This cap did not apply to taxes, insurance, and other payments, however.

⁴⁴Such loans typically would be sold privately at a significant discount. The GSEs and FHA took some steps to limit pipeline risk during the pandemic, but these steps only partially protected lenders (e.g., the GSEs agreed in April 2020 to purchase loans in forbearance, but only at a 500 to 700 bp discount (ABA, 2020)).

rising default risk is priced into mortgage rates, we should therefore see an increase in the interest rate *spread* between low-score and high-score conforming loans.

We first estimate this default risk premium using Optimal Blue Insight data, which include offer rates for otherwise identical mortgages with a FICO score of 680 and 750. We measure the rate premium for the FICO 680 loan by estimating the regression:

$$\text{rate}_{imt} = \alpha_{mt} + \beta_t \times (\text{FICO}_i = 680) + \varepsilon_{imt}, \quad (4)$$

where rate_{imt} is the interest rate on an offer i in CBSA m during week t , $\text{FICO}_i = 680$ is a dummy for a FICO score of 680 rather than 750, and α_{mt} is a set of CBSA $\times$ week fixed effects. Other loan characteristics are held fixed (e.g., DTI=36, LTV=80 and balance of \$300K). β_t then traces out the evolution of the FICO 680 to 750 interest rate spread.

We also compute an analogous spread using Optimal Blue rate locks data, estimating:

$$\text{rate}_{ilmt} = \alpha_{mt} + \delta_{lt} + \beta_t \times \text{FICO bin}_i + \Gamma X_{ilmt} + \varepsilon_{ilmt}, \quad (5)$$

where rate_{ilmt} is the interest rate on lock i issued by lender l in CBSA m during week t ; FICO bin_i includes five FICO bin dummies; α_{mt} and δ_{lt} are CBSA $\times$ week and lender $\times$ month fixed effects; and X_{ilmt} is a set of controls including log loan amount, DTI, DTI², and dummies for lock period and property type. For loans with discount points or credits, we convert the lock rate to a zero-points equivalent interest rate using the market rate-point trade-off measured each week in Optimal Blue Insight.⁴⁵

Estimates of β_t from these two models are reported in panels A and B of [Figure 9](#). We find no evidence of a persistent increase in the rate spread on low-FICO loans for either offers or rate locks. The spread did spike temporarily in March 2020, but by April

⁴⁵The procedure is detailed in footnote 14. An alternative approach is to directly control for points and credits interacted with time dummies in equation 5; this generates similar results.

it returned to a narrow band around 40bp, at or even slightly below pre-pandemic levels. **Figure 9**

Data on quantities paint a similar picture. Panel C of **Figure 9** plots the number of **about** conforming lenders that offered mortgages to borrowers with different FICO scores, **here]** calculated by averaging the number of rate offers in the Optimal Blue Insight data across the 20 Case-Shiller metropolitan statistical areas. The number of lenders drops temporarily in March, and more so for lower FICO loans, but it is subsequently quite stable, with little evidence of rationing to riskier borrowers. Panel D plots the percentage of conforming rate locks below two FICO thresholds (680 and 640). For purchase loans (the two solid lines), there is no evidence of a drop in the fraction of low-FICO rate locks. There is a drop for refinances, but this is typical during a refinancing boom because high-FICO borrowers refinance more quickly when rates fall (Keys et al., 2016).

To sum up, we find heightened default risk was not an important driver of high markups on conventional conforming loans during the pandemic. Default risk was, however, more significant for rates in the high-risk FHA market, as we discuss in Section V.

C. *Macroeconomic and Health Shocks*

Although we find little role for individual default risk, perhaps mortgage rates incorporated a more general risk premium due to the direct macroeconomic and public health effects of the virus. In this case, we may expect to see heterogeneity in rates by *location*, depending on the severity of the pandemic and the drop in economic activity. Recall from **Figure 7** that while mortgage rates do vary across metro areas, the differences are typically small, and the change in interest rates after the pandemic began was very similar across metros. This suggests there is limited scope for local factors. Nevertheless, we now study whether the variation that does exist is linked to local macroeconomic and health shocks.

In [Table VI](#), we examine variation in locked rates across the 100 most populous metro areas, focusing on the early stages of the pandemic when uncertainty and the economic effects of the virus were most acute. We regress interest rates (again, adjusted for points) from 1.1 million conventional conforming loans locked from November 2019 through August 2020 on CBSA fixed effects, loan characteristics interacted with week locked, and different proxies for how severely a metro area was affected by the health and economic effects of COVID-19.

Table VI

The first variable we try is COVID-19 cases per thousand residents in the prior calendar month. In fact, a higher case rate is associated with slightly *lower* mortgage interest rates (column (1)), although the effect is extremely small—a one-standard-deviation rise in the case rate is associated with only a 0.3bp drop in mortgage rates. Results are similar if we instead use a dummy for metros in the top quartile of cases per capita (column (2)).

about

here]

Next, we study the economic shock. Job losses, measured as the standardized year-over-year change in the local unemployment rate, are positively correlated with mortgage rates (column (3)), but again the effect is not economically meaningful. A 1 std increase in unemployment is associated with a 1.1bp rise in mortgage rates. The national increase in unemployment (from 3.5% to a peak of 14.8%) corresponds to 2.7 standard deviations, or a rate impact of only 3bp. Alternatively, we use a measure of time spent at workplaces (from the Opportunity Insights Tracker, Chetty et al., [2024](#)). One might expect that interest rates were higher in places where workplaces were shuttered, because of higher risk or difficulty in closing loans; directionally, however, we find the opposite, although the effect is again small, as seen in column (4). (Note: the sample size in this column is somewhat lower because “time-in-workplaces” is not available for all CBSAs in our sample.)

To summarize, we find no evidence that cross-metro variation in the spread of COVID and the economic effects of the virus was meaningfully priced into mortgage rates.

D. Market Power and Shopping

Finally, we investigate whether lender market power can account for the rise in intermediation markups. Market power may arise through concentration; for example, Scharfstein and Sunderam (2016) find evidence of lower mortgage rate pass-through in concentrated mortgage markets. Alternatively, even if there are many lenders, each originator may enjoy market power if borrowers face switching costs or do not search extensively (Klemperer, 1987; Wolinsky, 1986). For example, it is possible that borrowers searched less actively once the boom began because they did not need to scour the marketplace to beat their current rate (see Bhutta et al., 2024 for related evidence).

We study the role of market power in several ways. First, following Scharfstein and Sunderam (2016), Table VI tests whether mortgage rates fell by less in concentrated markets, using two ex-ante concentration measures derived from HMDA data interacted with a COVID-19 dummy (= 1 from March 14, 2020). In column (5), the concentration measure is the share of loans in 2019 originated by the metro area’s top four lenders, while in column (6) it is the local Herfindahl-Hirschman Index (HHI). We find no evidence that concentration mattered for rate pass-through—neither interaction term is statistically significant, and the confidence bounds exclude an economically significant effect.

Second, we repeat this exercise using the market share of lenders that failed during the 2007 to 2008 crisis, which Buchak and Jørring (2024) argue provides plausibly exogenous variation in post-crisis local mortgage market competition.⁴⁶ We find a correctly signed

⁴⁶We thank Adam Jørring for providing us with a copy of this variable.

but economically minor effect (column (7) of [Table VI](#)); a 1 std increase in the failed lender share is associated with a 0.8bp larger fall in mortgage rates.

Third, we examine whether market concentration increased during the pandemic, measured by HHI or by the number of unique active lenders at the CBSA-by-month level (see [Section V.A](#) of the Internet Appendix). By either metric, local markets in fact became *less* concentrated, consistent with our evidence from [Section III.C](#) that lenders expanded to new markets as demand rose. In short, we find no evidence that the pandemic somehow limited the set of lenders that borrowers could access.

Fourth, we use NSMO data to study whether borrowers searched less actively or changed their shopping behavior during the pandemic, considering six outcomes such as whether the borrower seriously considered more than one lender or made use of many information sources when searching. Results are presented and discussed in [Section V.B](#) of the Internet Appendix. To summarize, we find no evidence that borrowers searched less actively, although there is some evidence the pandemic led borrowers to put lower value on having a prior relationship with their lender, a shift that, if anything, would enhance competition.

Fifth, we use Google Trends data to study online mortgage search activity ([Figure IA.17](#) in the Internet Appendix). Mortgage search volume spiked to record levels in March 2020, and remained high through 2021. Further, the figure shows that search activity was unusually high relative to what would be predicted by the level of refinancing incentives.

In sum, we find no evidence that lender market power can explain the sharp increase in markups—borrowers were if anything searching unusually actively, local markets became less concentrated after the boom began, and interest rate pass-through was not closely

connected to measures of local competition. Finally, Black Knight (2020) shows that servicer retention—the share of borrowers refinancing through their existing servicer—fell during the pandemic to only 18% by 2020:Q3, the lowest rate in at least 15 years.

V. Credit Supply in Riskier Market Segments

Finally, we turn from conventional conforming mortgages to study credit supply in two other segments—the jumbo market and the FHA market. The jumbo market is a useful laboratory because it does not feature government guarantees and did not benefit directly from Fed QE. By comparing jumbo loans to conforming and “superconforming” loans, we estimate how government guarantees and QE separately played significant roles in supporting mortgage supply. That said, we also show that guarantees were not enough to fully stabilize lending in the FHA market, where credit availability also declined relative to the prime conforming market. Our key findings are summarized below. Full estimation results and further discussion are presented in Section VII of the Internet Appendix.

A. *The Jumbo Market*

Analyzing Optimal Blue data using the same methodology as in Section IV.B, we find that the interest rate spread between prime jumbo loans and otherwise identical conforming loans increased sharply by 40 to 70 bp at the start of the pandemic before gradually normalizing over the last few months of 2020 (see Section VII.A of the Internet Appendix). The number of lenders offering jumbo loans also dropped by more than half for prime borrowers, and fell to essentially zero for risky borrowers. The quantity of credit as measured by the share of jumbos in total lock volume also decreased. We confirm these patterns in HMDA data, finding that the share of jumbo originations declined by 30% to

50% during the pandemic (measured within a narrow 10% window on either side of the conforming loan limit).

This decline in credit availability in the jumbo market compared to the conforming market is consistent with an amplification of credit risk premia, given that jumbos do not carry government credit guarantees. But an additional factor is that conforming loans (but not jumbos) were purchased in large quantities by the Fed starting in March 2020 through its QE program.⁴⁷ To disentangle the effects of QE and government guarantees, we study the supply of *superconforming* mortgages. These are loans in high-cost areas with balances below the local conforming limit—making them therefore eligible for government guarantees—but above the national conforming limit, which for institutional reasons makes them much less likely to be purchased by the Fed.⁴⁸

Section VII.A of the Internet Appendix shows that the superconforming-conforming spread increased by about 25bp after Fed MBS purchases restarted in March, consistent

⁴⁷The Fed purchased \$580bn in agency MBS through the TBA market in March and April 2020 alone (Frame et al., 2021), and its agency MBS holdings increased rapidly from \$1.37tr at the start of March 2020 to \$1.90tr by the end of June (source: Federal Reserve Bank of New York). The pace of purchases slowed after April 2020 but purchases continued until 2022. Importantly, Fed QE did not include purchases of nonagency MBS backed by jumbo loans. (To date, the Fed’s long-term asset purchases have been restricted to credit-insensitive assets. The Fed did purchase corporate bonds in 2020, although this was done over a shorter period under the Fed’s Section 13(3) emergency lending authority; see e.g., Gilchrist et al., 2024.)

⁴⁸Specifically, the Fed buys agency MBS in the “to-be-announced” or TBA market, and pools comprising more than 10% of superconforming loans are not TBA eligible (Vickery and Wright, 2013; Huh and Kim, 2021). Matching data on the Fed portfolio holdings to eMBS loan-level data, we estimate in Section VII.A of the Internet Appendix that superconforming loans are about 40% less likely to be purchased by the Fed.

with a QE channel. The effect dissipated by June, however. Studying quantities, we find that both QE and government guarantees seem to have promoted credit supply—lending dropped in relative terms just above *both* the national and local conforming limits during the pandemic (restricting our analysis to high-cost areas where the two limits are different). The estimates around the local limit are more persistent and larger relative to the sample mean, however, suggesting that credit guarantees had more significant effects. That said, our results provide new evidence that QE has “local” effects on lending—even within the conforming market, QE had larger supply effects for the specific mortgages likely to be purchased by the Fed.⁴⁹

B. The FHA Market

While credit guarantees and QE were important, we find they were not enough to fully stabilize credit in the FHA market. Again using Optimal Blue data, Appendix VII.B finds a spike in the interest rate spread between higher-vs.-lower risk FHA loans starting in April 2020 and peaking in June, coincident with the post-CARES-Act surge in forbearance. The number of lenders offering FHA loans also fell significantly in April, and the share of FHA purchase rate locks to low-FICO borrowers dropped by half. Intermediation markups also increased more rapidly in the FHA market. Section VII.B of the Internet Appendix shows that the primary-secondary spread widened by 40 to 60 bp more for FHA loans than conforming loans early in the pandemic; gain-on-sale also rose by more.

⁴⁹Here, we extend prior work showing that QE boosts conforming mortgage lending relative to jumbo lending (e.g., Di Maggio et al., 2020). An advantage of our approach, which focuses on variation in QE within the conforming market, is that jumbo and conforming loans also differ in other ways (e.g., the former are not guaranteed, and have higher risk-weights for banks since they cannot be securitized into agency MBS).

These results are consistent with an amplification of the risk premium for FHA lending, which was then priced into mortgage rates. An alternative hypothesis, however, is that FHA lending was “crowded out” because capacity-constrained lenders wanted to focus on loans that were simpler to underwrite. While it is difficult to fully disentangle these two stories, we are able to shed some light by studying processing times for FHA loans and riskier conforming loans before and during the pandemic, as a proxy for processing complexity (see Section B.3 of the Internet Appendix). We show that FHA loans do take longer to originate, but that the same is true for low-FICO conforming loans, a segment where interest rate spreads did *not* widen relative to prime loans, as shown in Figure 9. This suggests that the increase in markups in the FHA market was not simply due to lenders adjusting their pricing to discourage applications from harder-to-process borrowers. Our results therefore provide some support for the view that policy interventions to limit the risks faced by FHA lenders (e.g., a Ginnie Mae liquidity facility as proposed by Kaul and Tozer, 2020) would help stabilize FHA lending during stress periods.

VI. Conclusion

The 2020 to 2021 pandemic period represents the most significant shock to U.S. mortgage lending since the global financial crisis, and studying this episode yields new lessons about the functioning of this key market. While lending boomed, we find that an increase in intermediation markups limited the pass-through of low rates to households due to capacity constraints amplified by pandemic-related operational and labor market frictions. We find new evidence based on this period that capacity constraints are national in scope and have not yet been significantly mitigated by online and digital lending technologies.

Nonbanks, which play an increasingly important role in the mortgage market, expanded lending more elastically than banks in 2020 to 2021, but remain reliant on securitization. Studying the role of public policy, we find that government guarantees support the flow of credit to riskier borrowers but are not always sufficient. QE also supports mortgage supply and has local effects based on which conforming loans are purchased.

Stepping back, it is striking that in 2020, intermediaries collected as much as 5% of a mortgage's balance for the service of linking borrowers with savers. Moreover, most mortgages during this period were simple refinances; while careful underwriting of such loans is sensible from an individual lender's perspective, rate-and-term refinances lower systemic default risk even if *no* effort is spent in underwriting. Meticulously underwriting rate-and-term refinances in a period with binding capacity constraints not only raises costs, but may also crowd out harder-to-originate loans. Our findings, therefore, reinforce arguments for a larger role for streamlined refinances and adjustable-rate mortgages, or alternative mortgage designs that feature automatic adjustments during periods of stress (e.g., Eberly and Krishnamurthy, [2014](#); Guren et al., [2021](#); Campbell et al., [2021](#)). As in the financial crisis, such contracts would have substantially strengthened the transmission of low interest rates during the pandemic.

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Figure 1. Mortgage rates, Treasury and MBS yields, and mortgage demand. Panel A displays time series of the 30-year fixed-rate mortgage (FRM) rate, the 10-year Treasury yield, and the MBS yield (based on interpolating between coupons, as explained in Section A.1). Panel B decomposes the spread between the 30-year FRM rate and the 10-year Treasury yield (shown by the solid line) into the MBS spread (= MBS yield minus 10-year Treasury yield) and the primary-secondary spread (= 30-year FRM rate minus MBS yield), as explained in Section II.A. Panel C shows market-wide mortgage application volume and the gap between the weighted average coupon (WAC) on outstanding mortgages and the 10-year Treasury yield (a proxy for refinancing demand). Data sources: Freddie Mac PMMS (30-year FRM Rate), J.P. Morgan Markets (MBS Yield, 10-year Treasury yield, WAC), Mortgage Bankers Association (MBA Application Index).

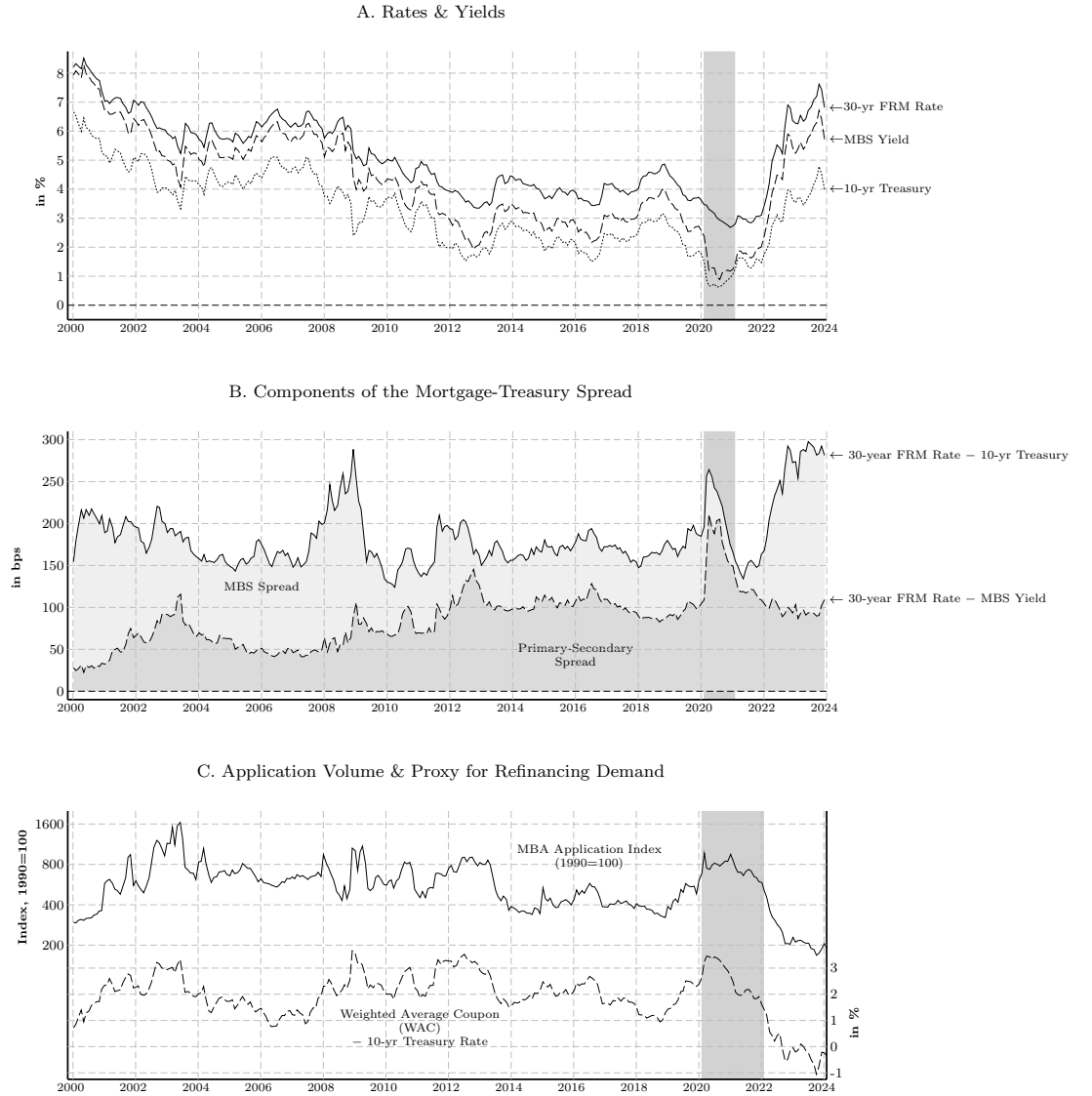


Figure 2. Tracing the mortgage-Treasury spread over 2020 to 2021. This figure decomposes the mortgage-Treasury spread into four components, based on the methodology described in Section II.A. The black dashed line shows the change in spread between the headline mortgage rate (from PMMS) and 10-year Treasury yields relative to the beginning of 2020, in basis points. Data sources: Freddie Mac PMMS, J.P. Morgan Markets.

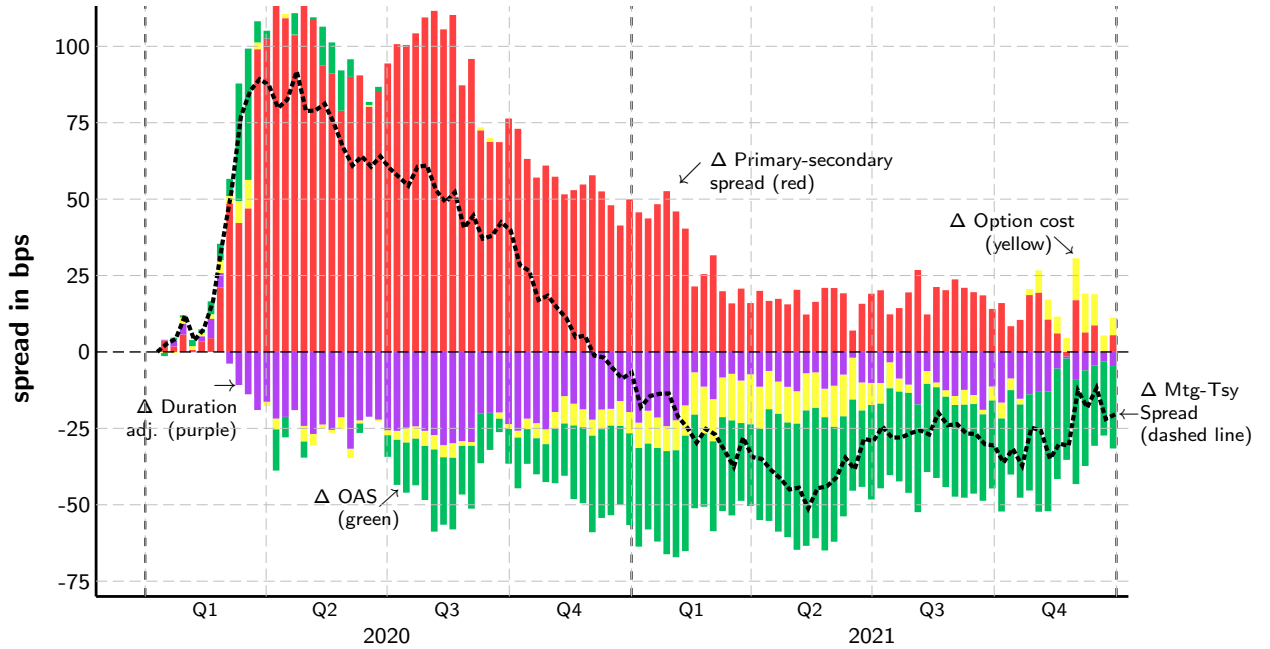


Figure 3. Measures of intermediation markups. This figure shows different measures of intermediation markups. The top left is the primary-secondary spread as explained in the text and the top right is gain-on-sale measured as described in Appendix I.A. Both measures are monthly averages constructed using weekly data from Freddie Mac and J.P. Morgan Markets. The bottom left panel uses measures from SEC filings for a selection of publicly-traded mortgage lenders reported in the “Mortgage Profitability: Production and Servicing” table in Inside Mortgage Finance (2023). Values are estimated quarter dummies from a regression of mortgage production income divided by total originations on quarter dummies and a full set of lender fixed effects. Bottom right panel is “Total Net Production Income, Basis Points, Simple Average” from Table B2 of the Mortgage Bankers Association Quarterly Performance Report (QPR).

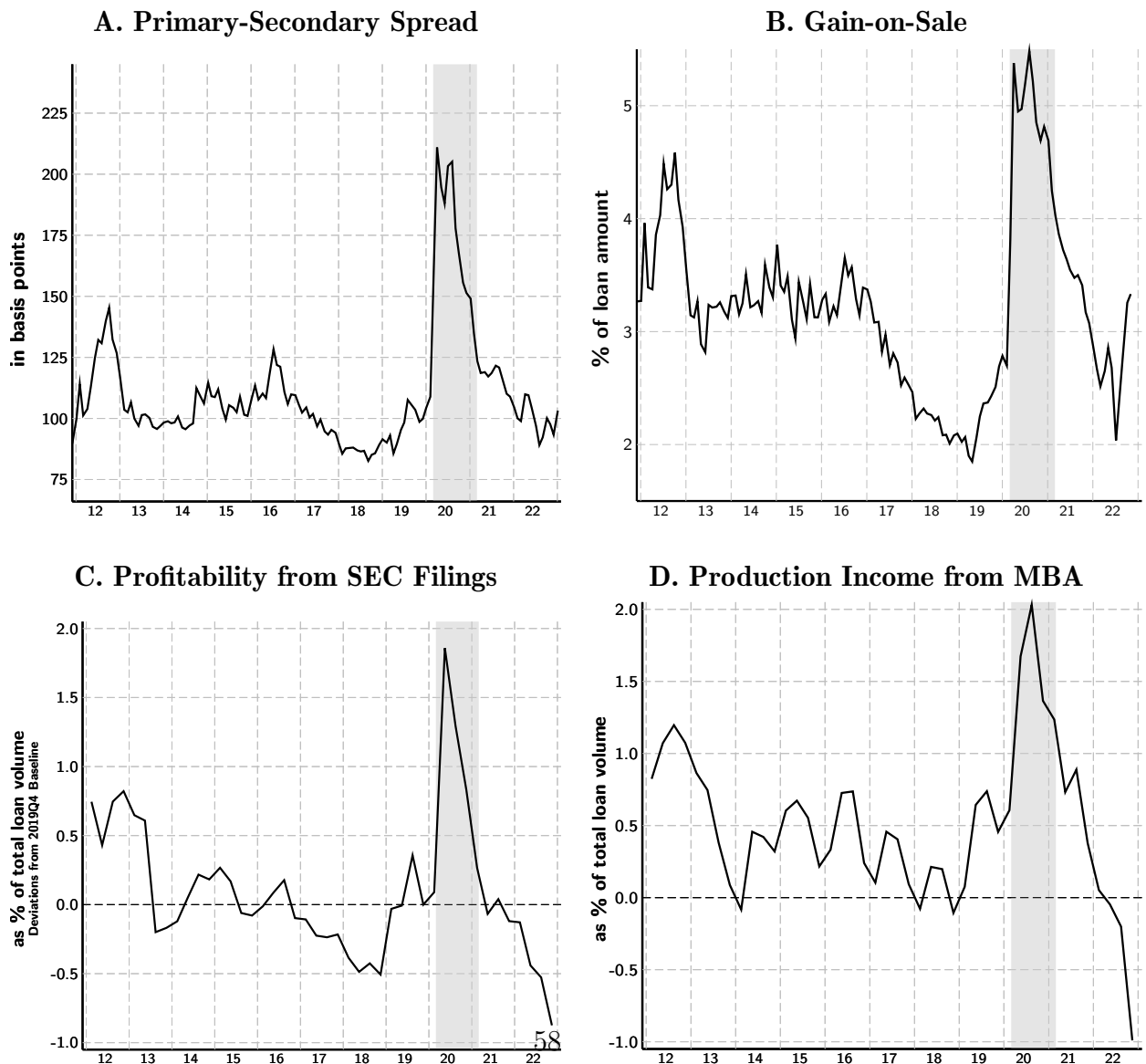


Figure 4. Intermediation markups and the demand for refinancing. Numbers next to red squares and green diamonds denote the calendar month in 2020 and 2021, respectively. The trend line and the 90% confidence intervals are estimates using data from 2012 to 2019. Spreads and gain-on-sale computed based on the methodology described in Section II.C. Data sources: Freddie Mac PMMS; J.P. Morgan Markets; SitusAMC; Mortgage Bankers Association (via Haver Analytics).

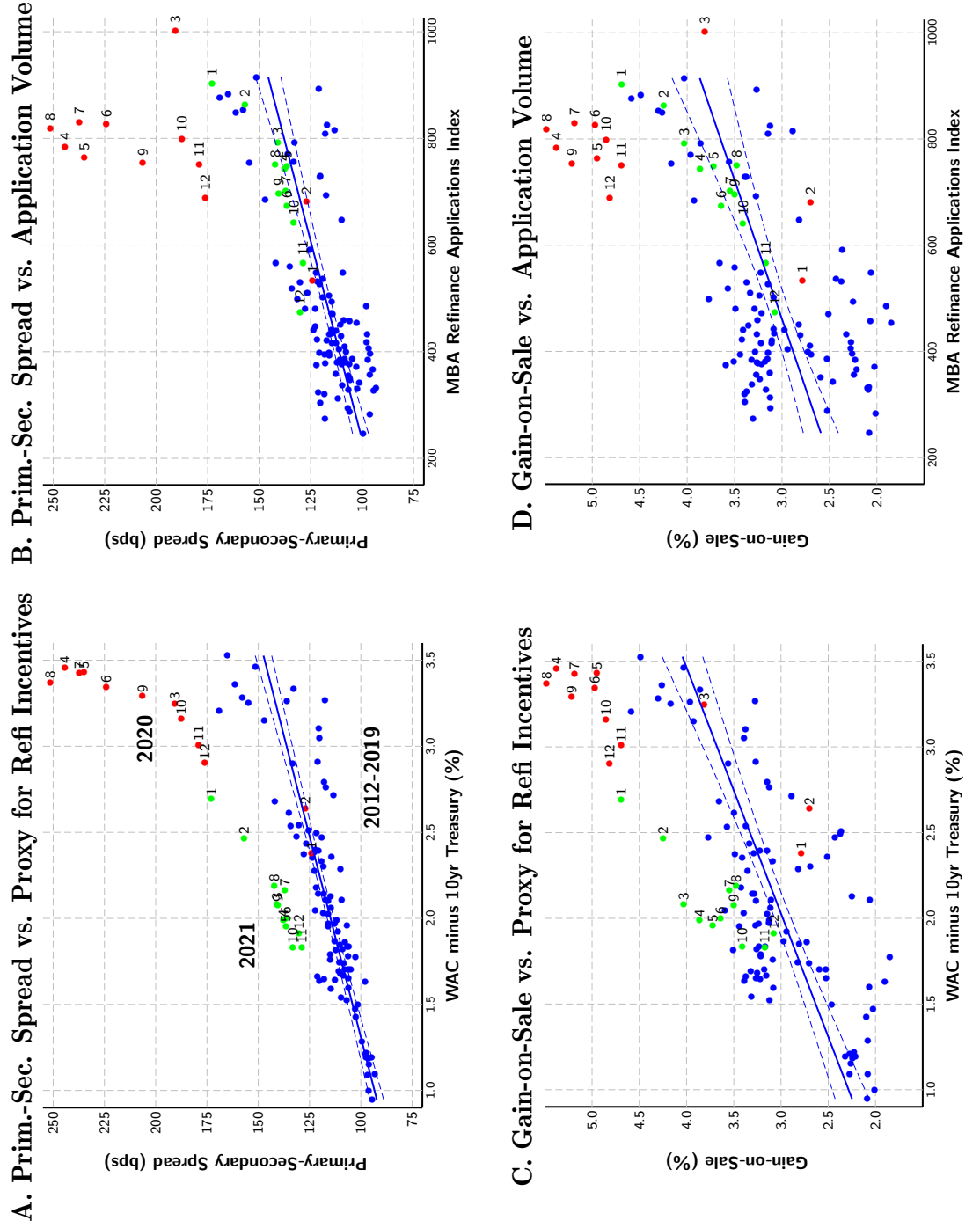
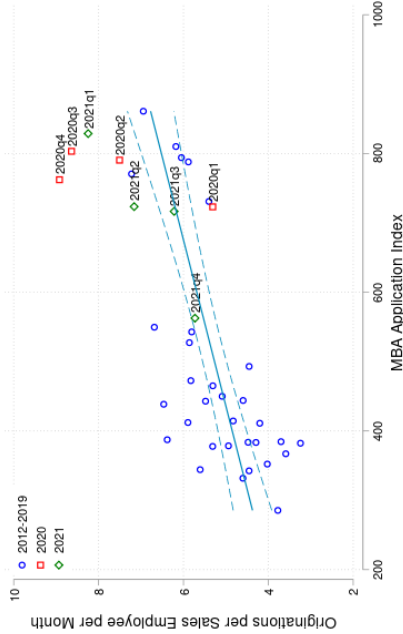
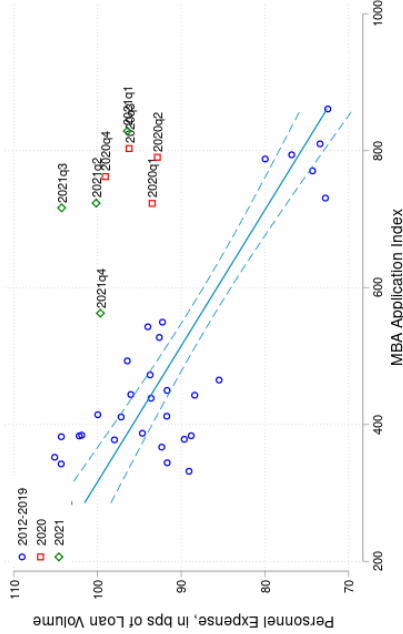


Figure 5. Loan officer production volume, personnel costs, job postings, and licensing. Panel A shows the mean monthly originations per sales employee at quarterly frequency from the Mortgage Bankers Association (MBA) Quarterly Performance Report (QPR). Panel B plots sales personnel expenses in basis points of loan volume, also from MBA's QPR. Both Panels A and B show the line of best fit from pre-pandemic observations over 2012 to 2019. Panel C plots the number of newly issued job postings for loan officers each month, sourced from Burning Glass. Panel D displays the number of newly licensed mortgage loan officers each month (by aggregating NMLS data), as well as the counterfactual number of licenses based on a monthly time-series regression of $\log(\text{licenses})$ on two lags of $\log(\text{applications})$ and a seasonal dummy for December.

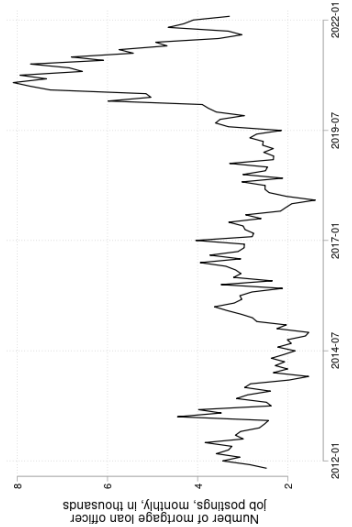
A. Originations Per Sales Employee



B. Personnel Costs (Sales Employees)



C. Job Postings for Loan Officers



D. New Loan Officer Licenses

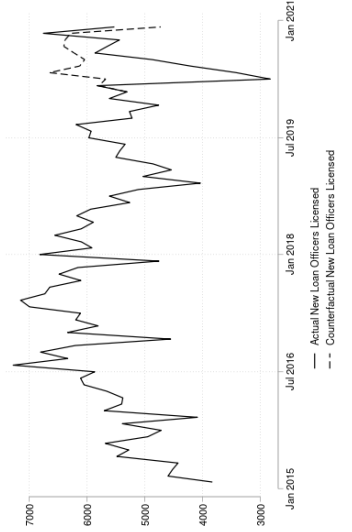
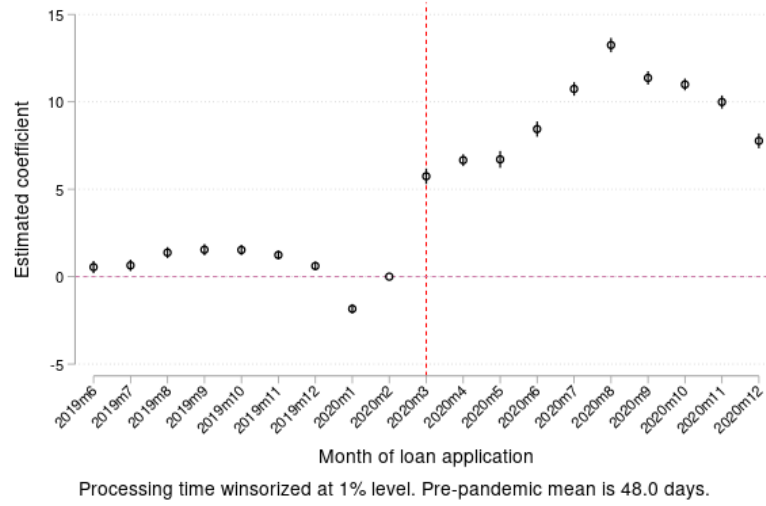


Figure 6. Delays in loan closings. Event study plots showing how mortgage processing times and the incidence of delays to loan closing evolved over 2019 to 2020. Panel A plots coefficients on time dummies from regression of mortgage processing time on application month dummies, loan and borrower characteristics, and CBSA dummies using confidential-use HMDA data. Panel B regresses a dummy for whether there were operational delays affecting loan closing on loan and borrower characteristics and origination month dummies using NSMO data.

A. HMDA: Processing Times (in Days)



B. NSMO: Experienced Issue(s) Delaying Loan Closing (0/1)

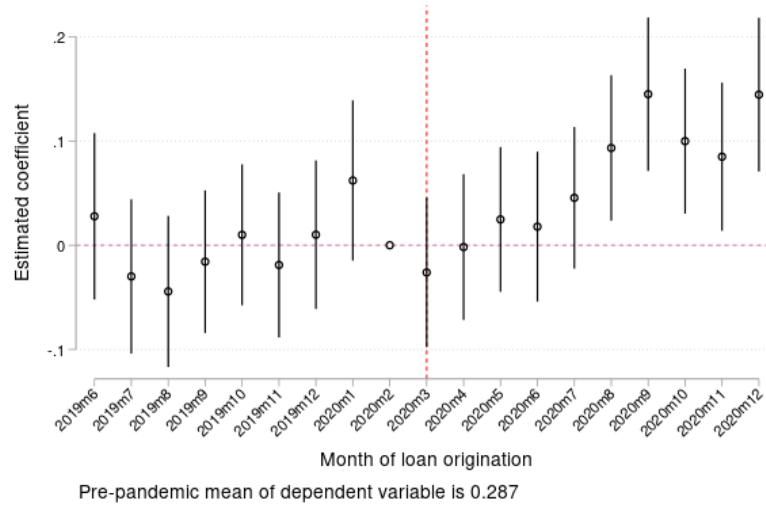
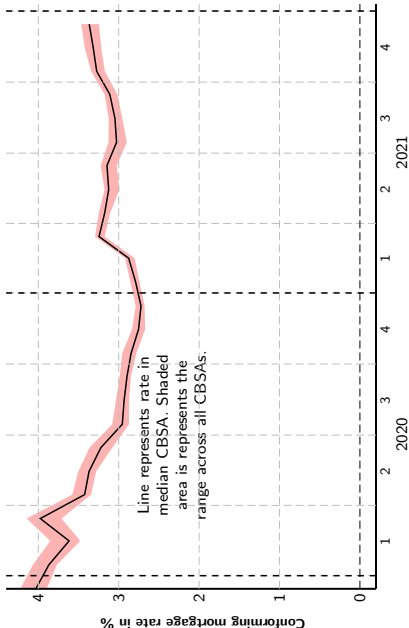
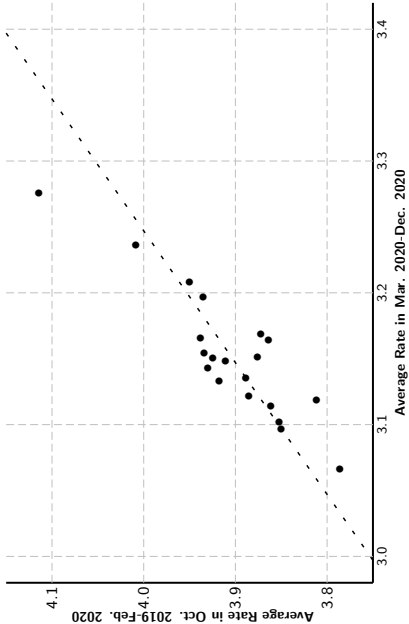


Figure 7. Dispersion in mortgage rates, application growth, and lender activity across metro areas. Panels A and B show rates in 20 CBSAs for conforming mortgages with loan amount = \$300k, LTV=80, FICO=750, DTI=36 from Optimal Blue Insight data. Panel A shows the time series movements for 20 CBSA. Panel B shows mortgage rates across these CBSAs averaged over the specified time periods. The dashed line in Panel B is a 45 degree line, estimated in a regression where the coefficient on pre-pandemic rates is forced to be 1. Panel C plots the interquartile range of year-over-year growth in mortgage applications measured at the CBSA level in HMDA data, weighted by the number of pre-period mortgage applications in each CBSA. Panel D plots the distribution of the mean and median number of CBSAs in which mortgage originators and mortgage loan officers in HMDA were active.

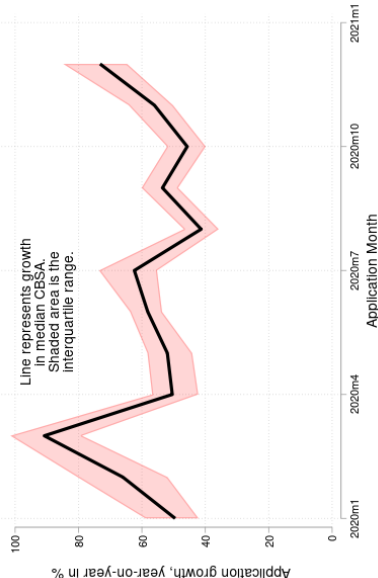
A. Time Series of Mortgage Rates by CBSA



B. Mortgage Rates by CBSA Averaged Over Different Periods



C. Year-over-year Growth in Mortgage Applications



D. Geographic Coverage

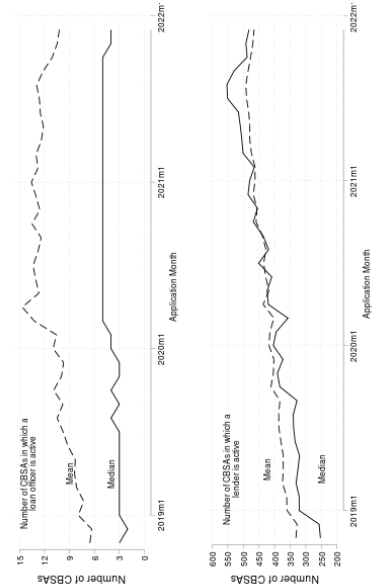


Figure 8. Nonbank share of conforming market. Nonbank share of conforming mortgage lending plotted against application date at a weekly frequency, constructed from confidential-use HMDA data. “No controls” plots the raw nonbank share of lending. “Only county FEs” plots the nonbank market share controlling for geography, estimated by regressing a nonbank dummy on time dummies and county dummies, then plotting the estimated time dummies. Similarly, “All controls” plots the nonbank share conditional on a larger set of controls, including dummies for refinancing and cash-out refinancing, log loan amount, log of applicant income, dummies for coapplicant, occupancy, pre-approval, applicant sex, race and ethnicity, DTI, DTI², credit score, credit score², LTV, LTV², bins of applicant age, county dummies, and dummies for missing values of each variable. Conforming loans are identified as mortgages that: i) do not exceed the relevant conforming loan limit and ii) are not flagged as government loans. Vertical dashed line indicates the onset of the pandemic, defined as the declaration of a national state of emergency on March 13, 2020.

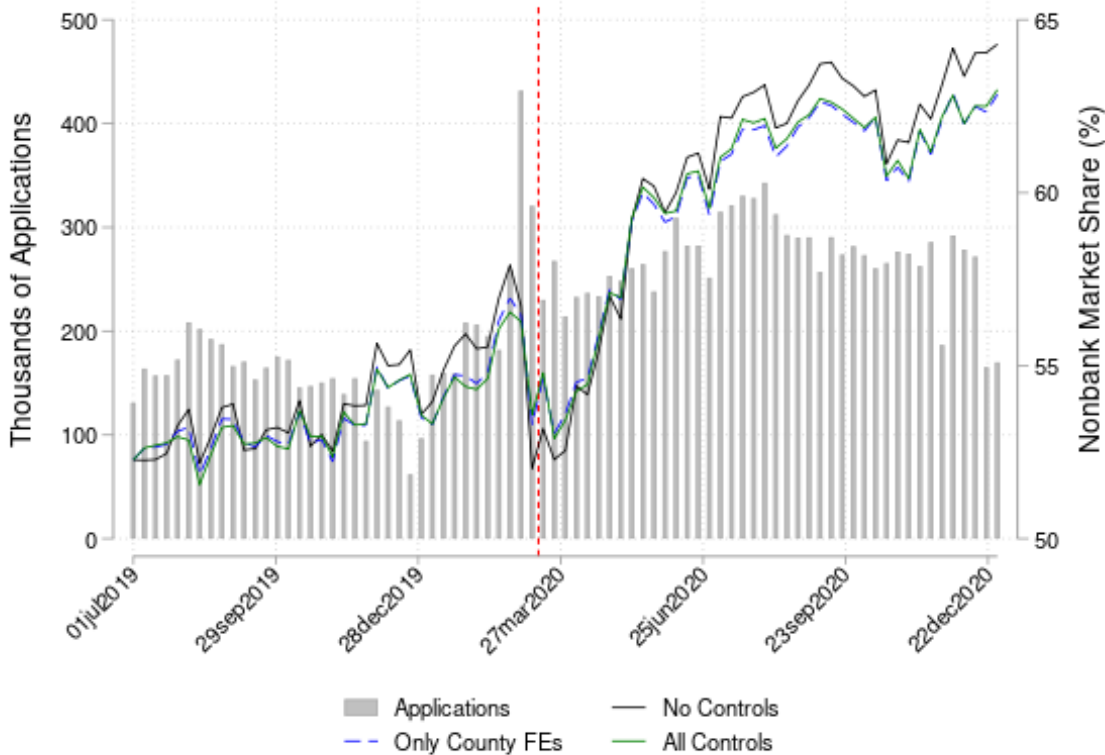


Figure 9. Default risk and credit supply in the conforming market. Panel A shows the spread between offered rates for conventional conforming borrowers with credit scores of 680 vs. 750. Panel B shows the spread in locked interest rates between those with scores of 680 vs at least 740. (See Section [IV.B](#) for details of methodology and controls.) Panel C displays the number of lenders each week posting offered rates for borrowers at or above particular credit score thresholds. Panel D displays the share of loans locked by borrowers at or below score thresholds. The data in Panels A and C come from Optimal Blue Insight, while Panels B and D use lock-level data from Optimal Blue. The vertical line in each panel represents the declaration of a national state of emergency on March 13, 2020.

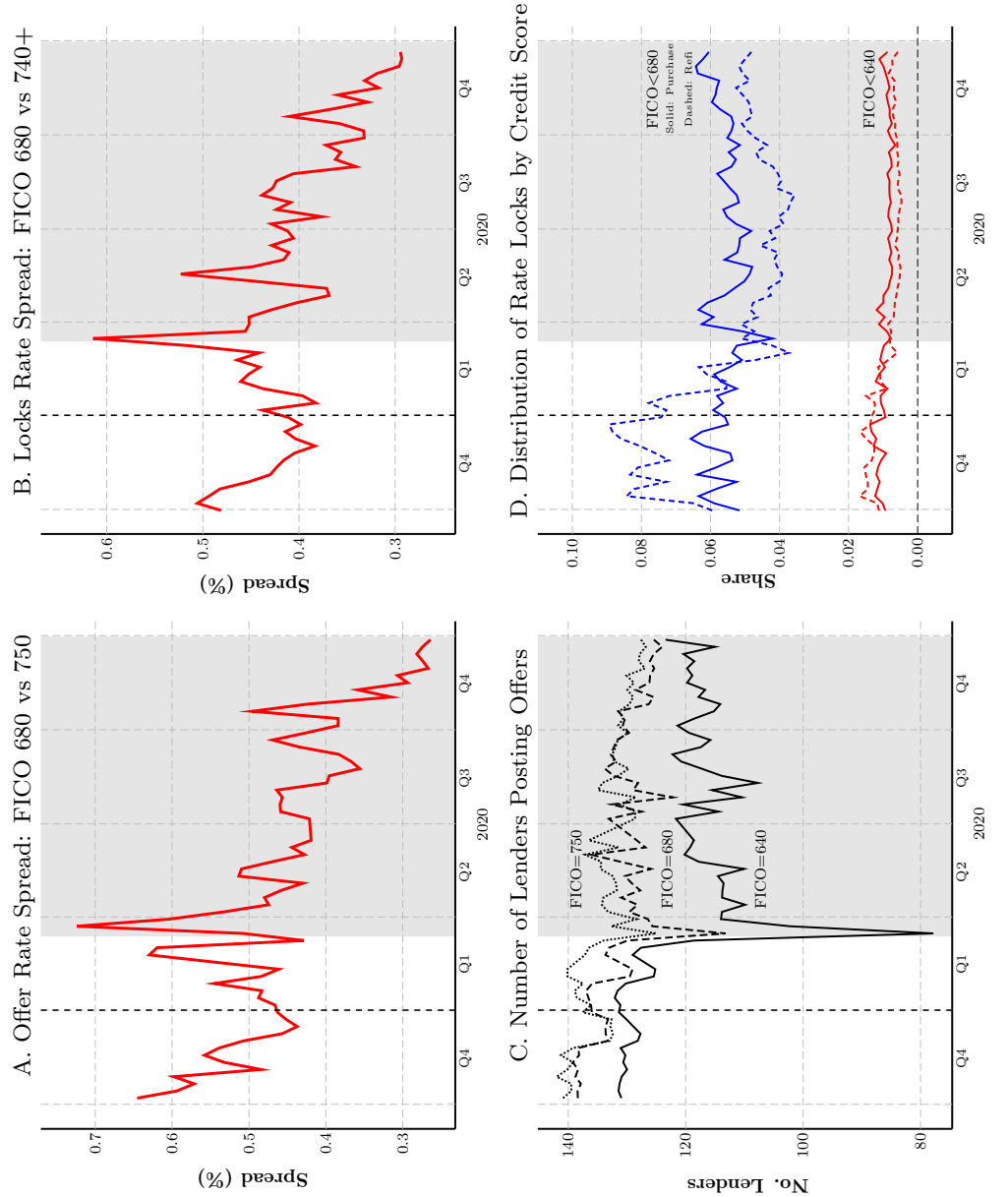


Table I. Intermediation Markups and Mortgage Demand. Data sources: Mortgage Bankers Association (MBA), J.P. Morgan Markets, and Freddie Mac 30-Year Fixed Rate Mortgage Average in the United States [MORTGAGE30US] and Board of Governors of the Federal Reserve System (US) 10-Year Treasury Constant Maturity Rate [DGS10], both retrieved from FRED, Federal Reserve Bank of St. Louis. MBA applications index is a three-week backward-looking moving average. Models include month-of-year dummies (coefficients not displayed). Observations are weekly and include January 2012 through December 2021. Newey-West standard errors (8 lags) in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Primary-Secondary Spread (bp)		Gain-on-Sale (\$ per \$100 face value)	
	(1)	(2)	(3)	(4)
Mortgage Demand:				
Refi incentive (WAC - 10 Year Treasury)	18.250*** (4.197)		0.441*** (0.097)	
MBA Applications Index		0.022** (0.010)		0.001* (0.000)
2020 Dummies:				
March – April	100.994*** (12.376)	121.212*** (13.348)	1.444*** (0.354)	1.830*** (0.405)
May – June	107.498*** (6.850)	130.712*** (5.985)	1.695*** (0.180)	2.154*** (0.183)
July – September	109.523*** (9.842)	129.245*** (9.639)	1.837*** (0.177)	2.210*** (0.199)
October – December	65.351*** (4.778)	77.607*** (5.143)	1.542*** (0.148)	1.742*** (0.184)
2021 Dummies:				
January – June	46.768*** (4.605)	46.821*** (5.732)	1.275*** (0.132)	1.187*** (0.178)
July – December	37.992*** (3.083)	35.178*** (3.897)	0.539*** (0.099)	0.398*** (0.140)
Constant	63.409*** (10.029)	89.384*** (7.443)	1.895*** (0.252)	2.453*** (0.251)
Num obs.	1190	1190	828	828
Mean of dep. var.	103.27	103.27	2.95	2.95
R ²	0.56	0.45	0.53	0.42
RMSE	24.18	26.90	0.59	0.65
RMSE (no dummies, -Feb. 2020)	20.88	26.91	0.59	0.70

Table II. Application Volume and Geographic Dispersion of Lending. Lender i (loan officer i)-by-month t models estimating the log number of distinct metropolitan areas in which a lender (loan officer) received applications over different phases of 2019 and 2020. Data source: confidential-use HMDA data. Time is indexed by application date. Sample period is January 2019 to December 2020. Omitted dummy is January–August 2019. Standard errors are clustered by lender in columns (1) to (3) and loan officer in (4) to (6). ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Lender-Level			Loan Officer-Level		
	(1) b/se	(2) b/se	(3) b/se	(4) b/se	(5) b/se	(6) b/se
ln(Total Applications Taken by Lender in Month t)	0.343*** (0.026)		0.367*** (0.021)	0.513*** (0.001)		0.500*** (0.001)
Sept. 2019 – Feb. 2020		0.014*** (0.005)	-0.008* (0.005)		0.008*** (0.001)	0.002** (0.001)
March – April 2020		0.178*** (0.017)	-0.050*** (0.012)		0.282*** (0.002)	0.012*** (0.001)
May – June 2020		0.170*** (0.018)	-0.025* (0.015)		0.252*** (0.002)	0.038*** (0.002)
July – Sept. 2020		0.207*** (0.020)	-0.028* (0.015)		0.299*** (0.002)	0.043*** (0.002)
Oct. – Dec. 2020		0.165*** (0.017)	-0.014 (0.016)		0.217*** (0.002)	0.051*** (0.001)
_cons	2.375*** (0.216)	5.120*** (0.010)	2.194*** (0.178)	0.067*** (0.003)	1.326*** (0.001)	0.081*** (0.003)
Num obs.	68408	68408	68408	3109860	3109860	3109860
Mean of dep. var.	5.23	5.23	5.23	1.48	1.48	1.48
SD of dep. var.	1.55	1.55	1.55	1.14	1.14	1.14

Table III. Fintech Lending. Loan-level linear probability models estimating conditional changes in the fintech share of conventional conforming mortgage lending during the pandemic as well as changes in the difference in mortgage processing time between fintechs and other mortgage lenders. Time is indexed by application date. Sample period is July 2019 to December 2020. Pandemic is defined as the period from March 2020 onwards. Loan controls include dummies for refinancing and cash-out refinancing, log loan amount, log of applicant income, dummies for coapplicant, occupancy, pre-approval, applicant sex, race and ethnicity, DTI, DTI², credit score, credit score², LTV, LTV², bins of applicant age, and dummies for missing values of each variable. Conforming loans are identified as mortgages that: i) do not exceed the relevant conforming loan limit and ii) are not flagged as government loans. Data source: confidential-use HMDA data. Standard errors clustered by county. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	=100 if lender is fintech; 0 otherwise				Processing time (days)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Pandemic	2.65*** (0.15)	1.64*** (0.17)	0.01 (0.14)	-0.18 (0.15)	10.12*** (0.13)	9.20*** (0.15)	9.31*** (0.15)
Pandemic x FICO<680				2.85*** (0.19)			
Fintech					-9.38*** (0.28)	-5.64*** (0.28)	-6.78*** (0.25)
Pandemic x Fintech					0.85*** (0.23)	1.77*** (0.21)	1.88*** (0.22)
Num obs.	13209832	7762129	7761406	7761406	13209832	7762129	7761406
Mean of dep. var.	16.06	27.34	27.34	27.34	54.65	50.70	50.69
Lenders	All	Nonbank	Nonbank	Nonbank	All	Nonbank	Nonbank
Loan controls	N	N	Y	Y	N	N	Y

Table IV. Nonbank Share of Lending. Loan-level linear probability models estimating conditional changes in the nonbank share of conforming mortgage lending over different phases of 2020. Dependent variable = 100 if a loan was originated by a nonbank mortgage company; = 0 otherwise. Time is indexed by application date. Sample period is December 2019 to December 2020. Omitted dummy is for the pre-period from December 2019 to February 15, 2020. Loan controls include dummies for refinancing and cash-out refinancing, log loan amount, log of applicant income, dummies for coapplicant, occupancy, pre-approval, applicant sex, race and ethnicity, DTI, DTI², credit score, credit score², LTV, LTV², bins of applicant age, and dummies for missing values of each variable. Conforming loans are identified as mortgages that: i) do not exceed the relevant conforming loan limit and ii) are not flagged as government loans. Data source: confidential-use HMDA data. Standard errors clustered by county. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
Early pandemic [Feb 15-Mar 12]	0.83*** (0.22)	1.50*** (0.15)	1.31*** (0.15)	-0.06 (0.30)	0.77*** (0.20)	2.78*** (0.17)
Nonbank stress [Mar 13-Apr 30]	-1.27*** (0.20)	0.25 (0.16)	0.11 (0.15)	-1.17*** (0.27)	-0.86*** (0.19)	-0.80*** (0.17)
Stress easing [May-Dec]	6.84*** (0.24)	6.72*** (0.20)	6.85*** (0.21)	5.23*** (0.22)	3.84*** (0.17)	7.02*** (0.23)
Num obs.	10605218	10605206	10602575	556634	1197757	8831743
Mean of dep. var.	59.85	59.85	59.86	57.41	60.42	51.81
Loan controls	N	N	Y	Y	Y	Y
County dummies	N	Y	Y	Y	Y	Y
Sample restrictions	none	none	none	CS<680	CLTV>90	no fintech

Table VI. Metro Area Differences in Conventional Conforming Mortgage Rates (%).

Mortgage lock-level regression of interest rate (adjusted for discount points and credits) on area characteristics and loan characteristics, including loans locked from November 2019 to August 2020 in the 100 largest CBSAs, with the exception of (4) and (9), which include locks from late February to August 2020, due to data availability. COVID cases per 1,000 are lagged one month. The top 4 lenders' market share, HHI, year-over-year unemployment, time spent at workplaces, and failed lender share are all standardized to have a mean of 0 and standard deviation of 1. Mortgage interest rates are adjusted for points paid by borrower (credits received from lender) using daily data from Optimal Blue Insight on offered interest rates for loans with different net points. Additional controls include CBSA-level fixed effects (county-level fixed effects in (4), (7), and (9)), as well as lock week interacted with: binned FICO score, binned loan-to-value ratio, interest rate type (fixed-rate, 5/1 ARM, 7/1 ARM, or 10/1 ARM), and loan purpose (purchase vs. refinance). Data Sources include Optimal Blue locks data, *New York Times* COVID data, Bureau of Labor Statistics unemployment data, confidential-use HMDA data, failed lender market share from Buchak and Jørring (2024), and county-level Google COVID-19 Community Mobility Reports data from the Opportunity Insights Tracker. Standard errors in parentheses, clustered at CBSA level for (1)–(3), (5)–(6), and (8) and at the county level for (4), (7), (9). ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
COVID cases per capita	-0.0032*** (0.0008)								
Dummy: MSA in top quartile		-0.0178*** (0.0051)						-0.0184*** (0.0050)	-0.0177*** (0.0039)
Year-over-year change in U.R.			0.0112** (0.0054)					0.0146*** (0.0049)	0.0157*** (0.0032)
Time at Workplaces				0.0217*** (0.0058)					0.0115** (0.0055)
COVID $\times$ top 4 share					0.0025 (0.0042)				
COVID $\times$ HHI						0.0048 (0.0042)			
1.post_covid#c.z.share.failed_2007							-0.0082*** (0.0023)	-0.0115*** (0.0028)	-0.0168*** (0.0027)
Num obs.	1130086	1130086	1130086	915233	1130086	1130086	1118397	1118398	905708
Mean of dep. var.	3.33	3.33	3.33	3.21	3.33	3.33	3.33	3.33	3.21